

Movelt2 Configuration

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Movelt2 is an advanced software framework for robot motion simulation, planning, and control.

Note: This configuration is pre-installed; please skip this step, otherwise it may damage your system environment!

1. Environment Setup

- Ubuntu 22.04 Virtual Machine
- ROS Humble

1.1 Installing Movelt2

```
sudo apt install ros-humble-moveit*
```

1.2 Creating a Folder

Create the automatically generated configuration file for Movelt2:

```
mkdir ~/dofbot_ws/src/dofbot_moveit
```

2. Movelt2 Configuration

Movelt Setup Assistant is a graphical tool used to simplify and accelerate the creation and configuration of Movelt2 configuration packages.

Using this tool, users can easily generate configuration files for robot motion planning and control without the need to manually edit complex configuration files.

2.1. Launching the Assistant

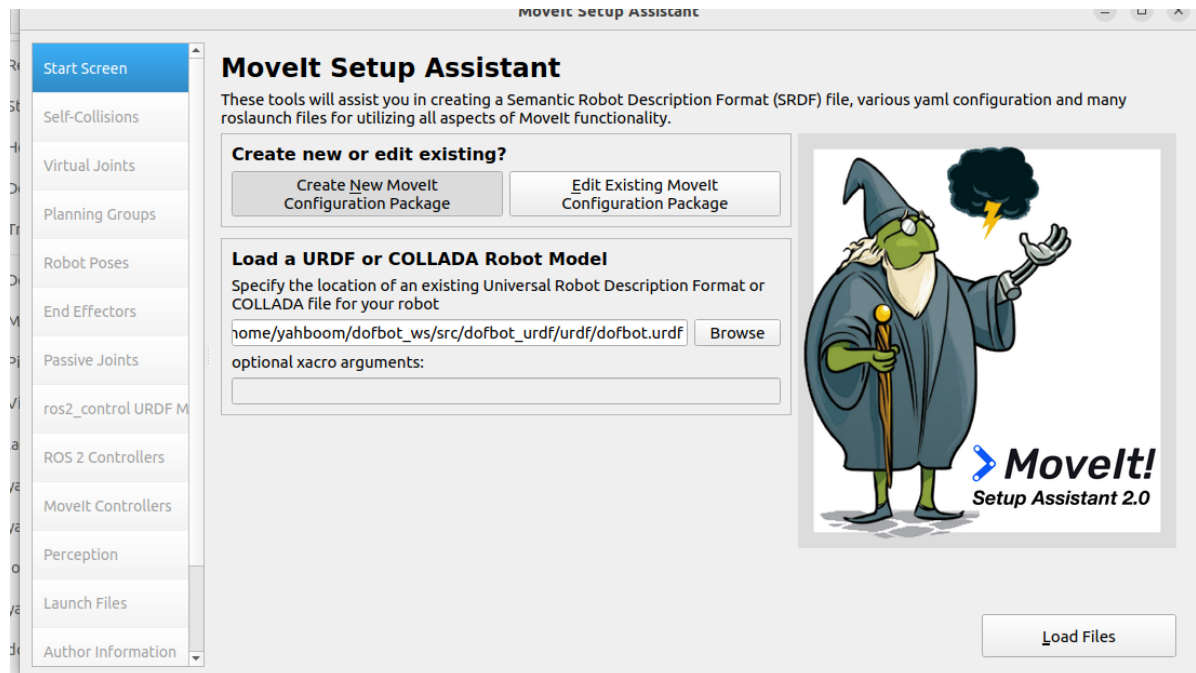
Launch the Movelt Setup Assistant:

```
ros2 launch moveit_setup_assistant setup_assistant.launch.py
```

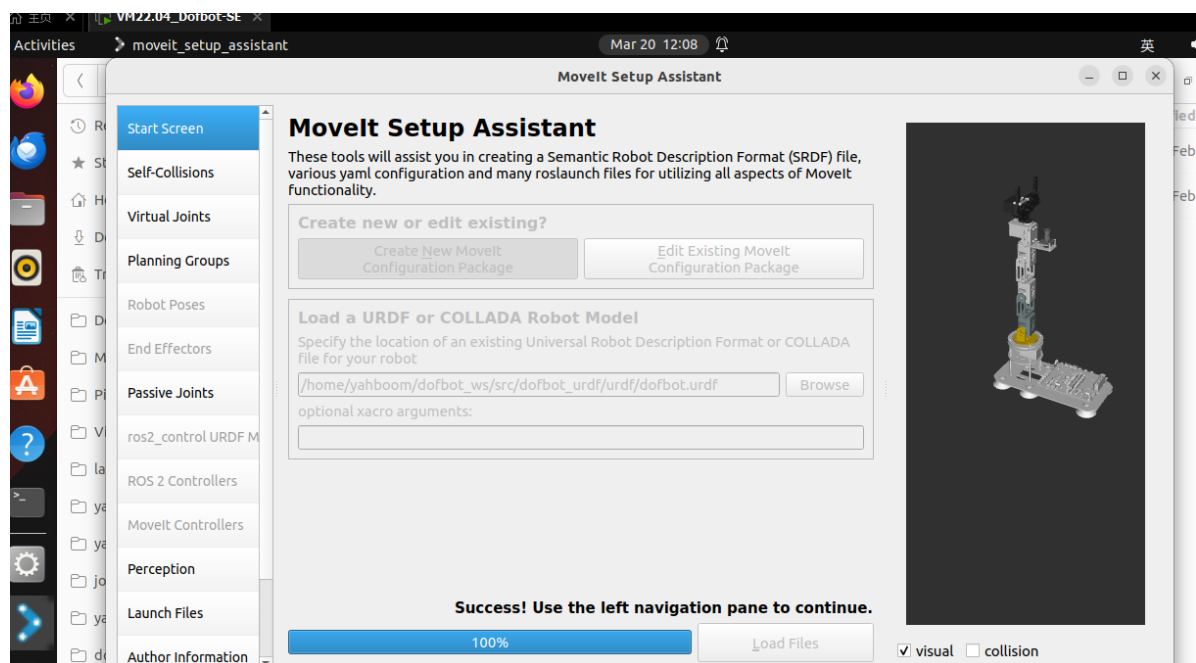
For first-time use, you can choose to create a new Movelt configuration package; for subsequent sessions involving only minor modifications, you can choose to edit an existing Movelt package.

Creating a new Movelt configuration package requires importing the robot's URDF model. Simply importing a standalone URDF model file into the Movelt Assistant will result in an error; instead, you must use the compiled package (specifying the path to the URDF file).

```
/home/yahboom/dofbot_ws/src/dofbot_urdf/urdf/dofbot.urdf
```



Once the URDF file has been selected, click "Load File," and the Movelt Assistant will display the robot model:

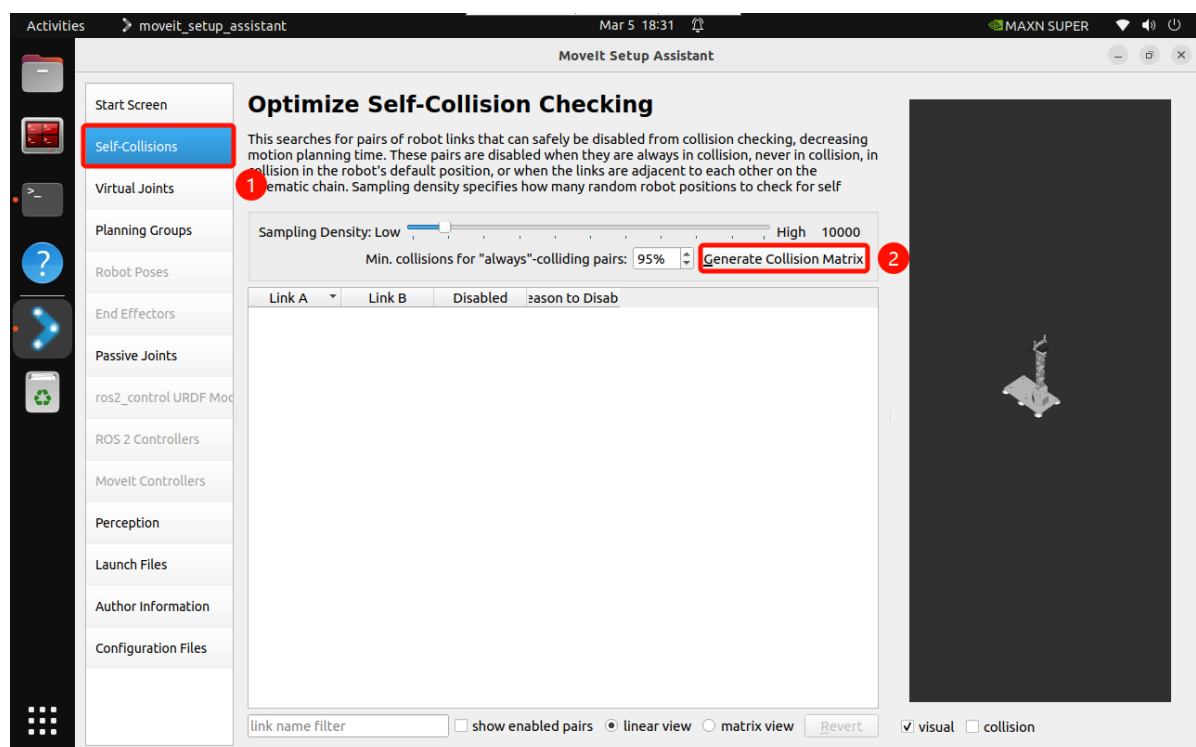


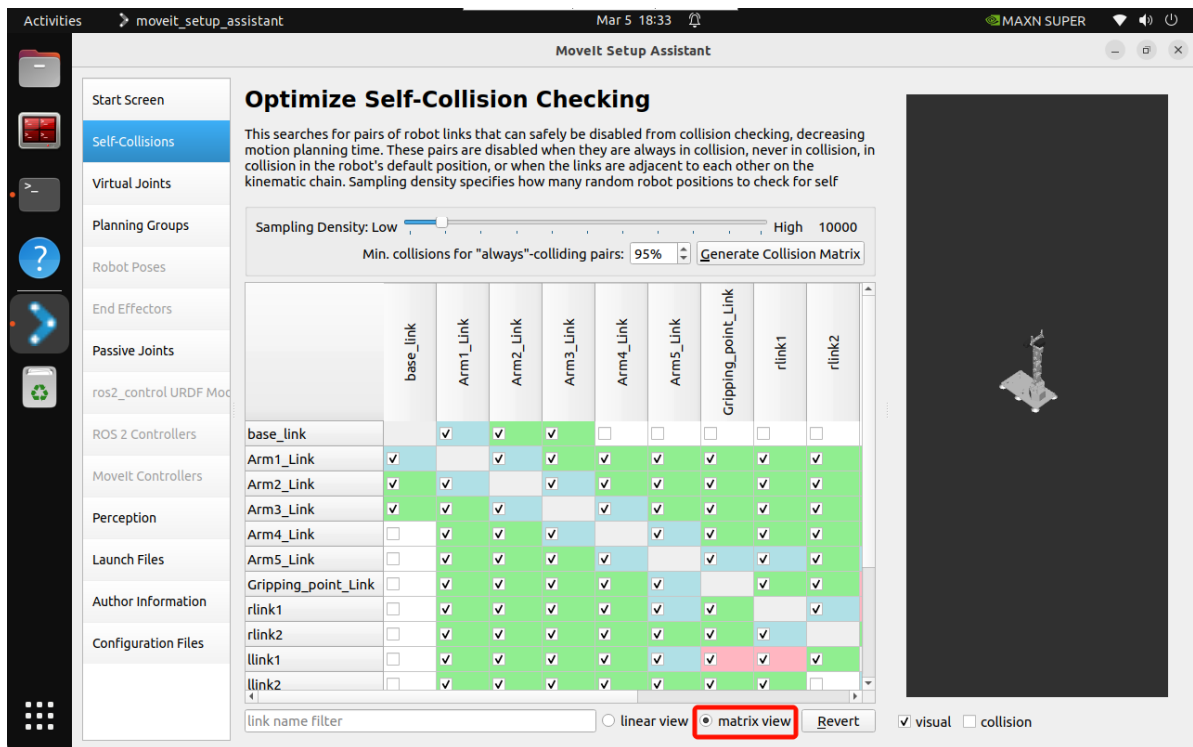
2.2. Collision Detection

The Self-Collision Matrix is a feature used to optimize motion planning. Its primary function is to generate a matrix that describes whether collisions are possible between the various links within the robot model.

This matrix helps the motion planner avoid unnecessary collision checks while planning paths, thereby improving planning efficiency.

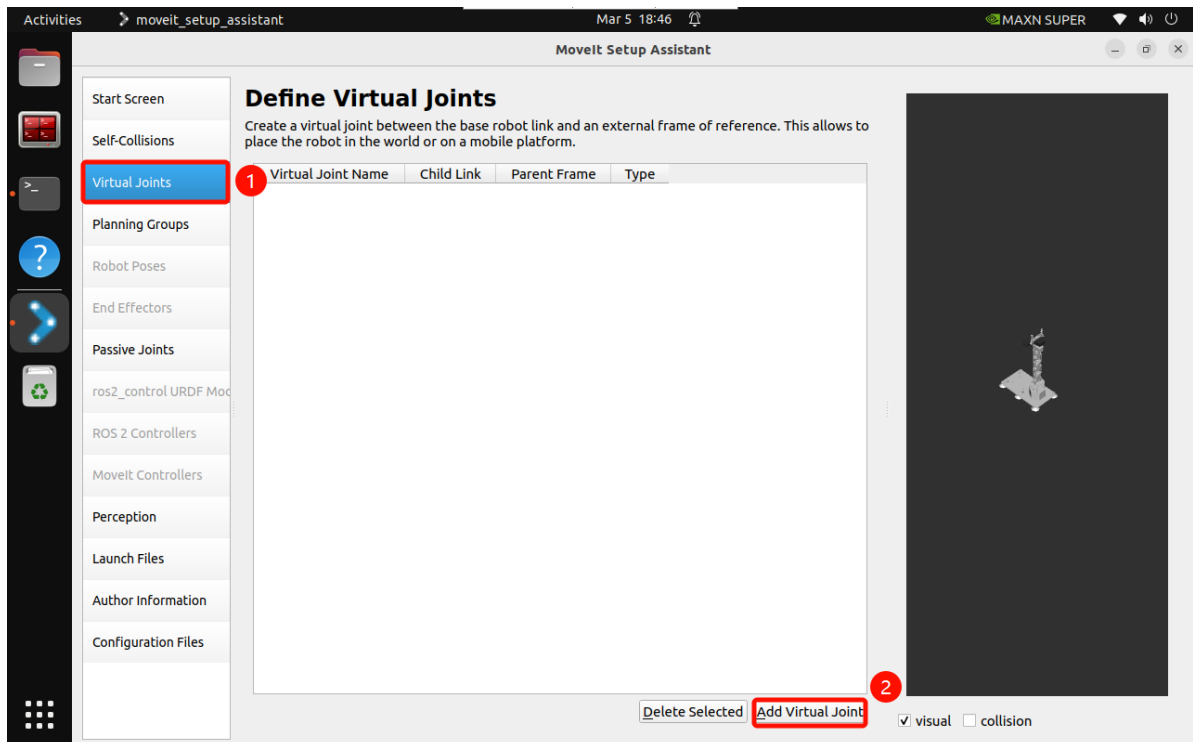
To generate the self-collision matrix, simply click the corresponding option within the MoveIt Setup Assistant to generate it automatically:





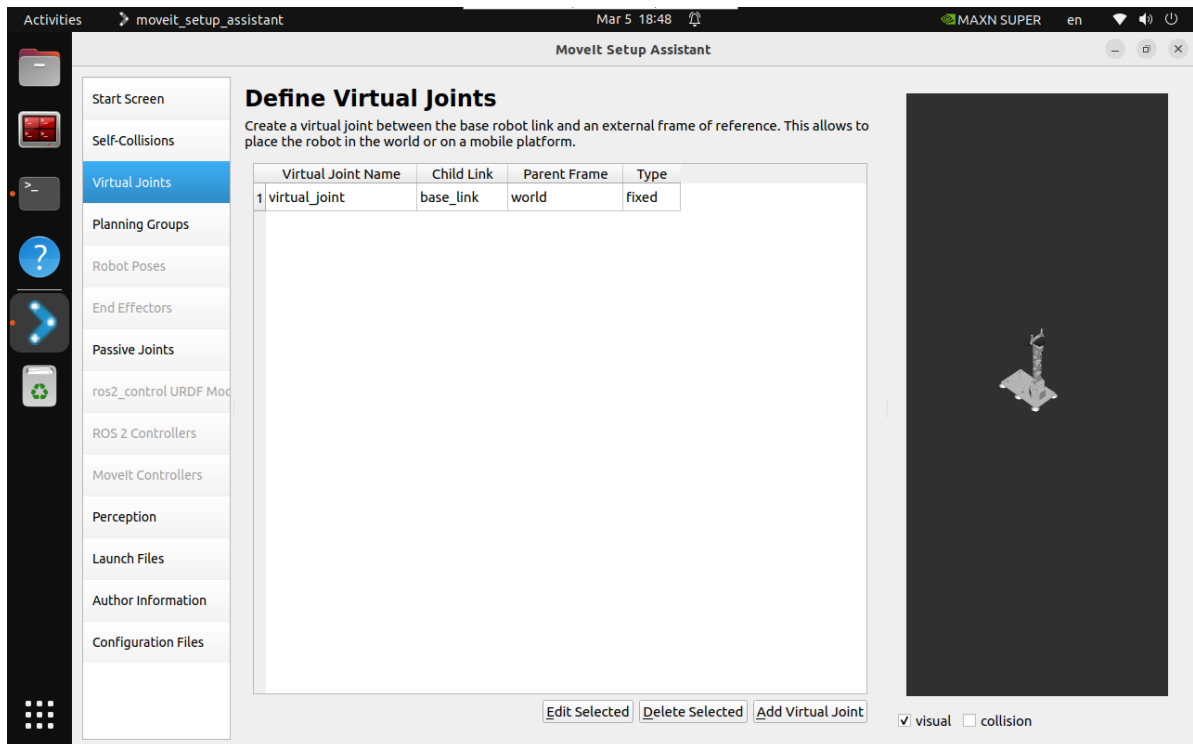
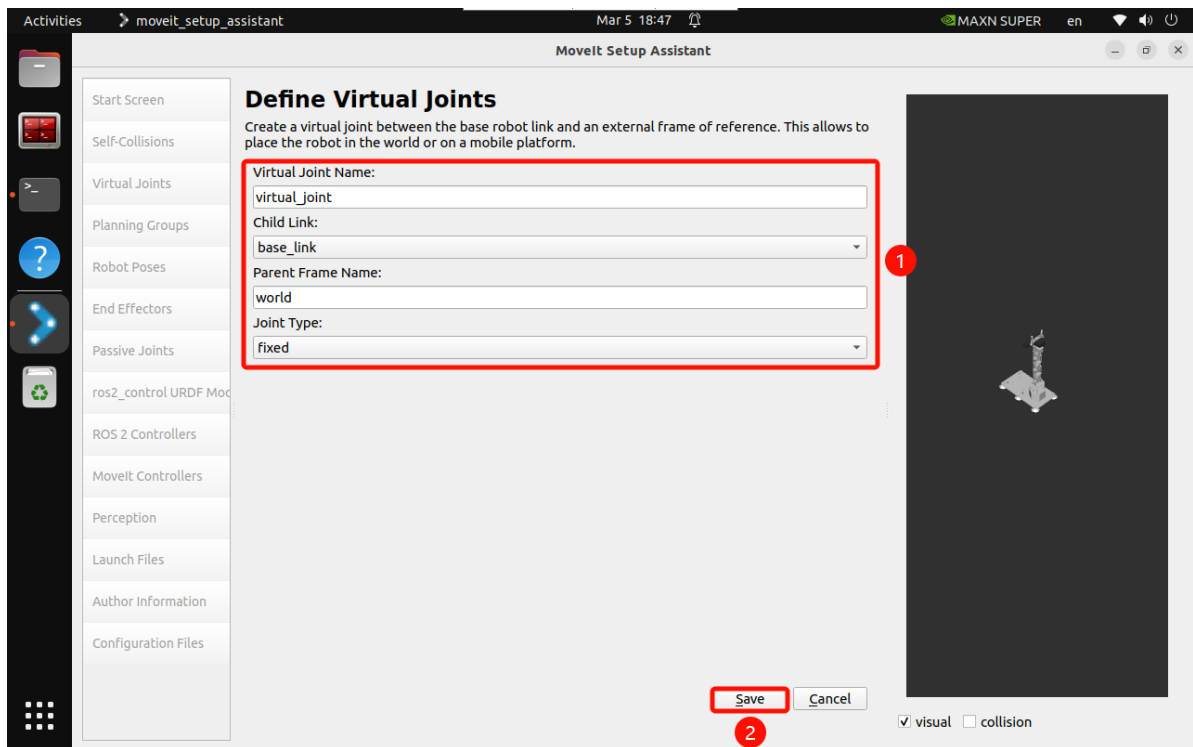
2.3. Virtual Joints

A virtual joint is used to define the relationship between the robot model and the external world. Virtual joints are typically used to describe the connection between a robot's base and a fixed reference frame (such as the world coordinate system).



To fix the robot's base link (`base_link`) relative to the `world` frame:

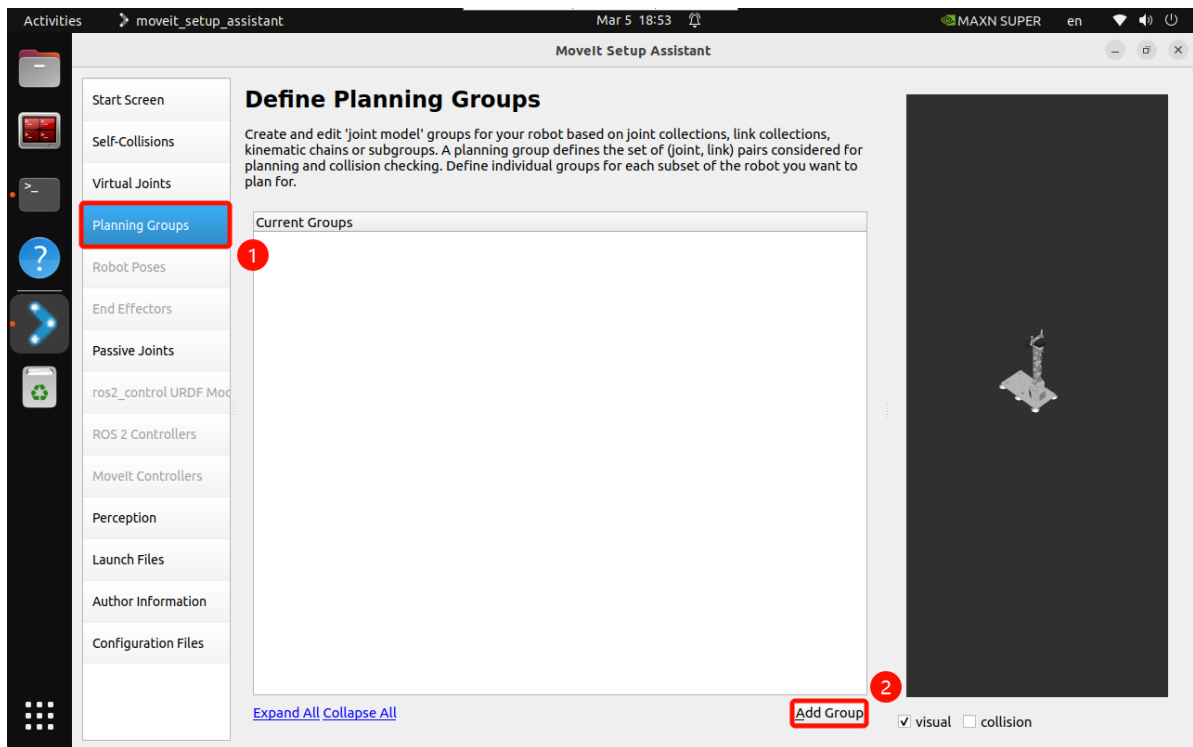
- Virtual Joint Name: `virtual_joint`
- Child Link: `base_link`
- Parent Frame Name: `world`
- Joint Type: `fixed`



2.4. Motion Planning Groups

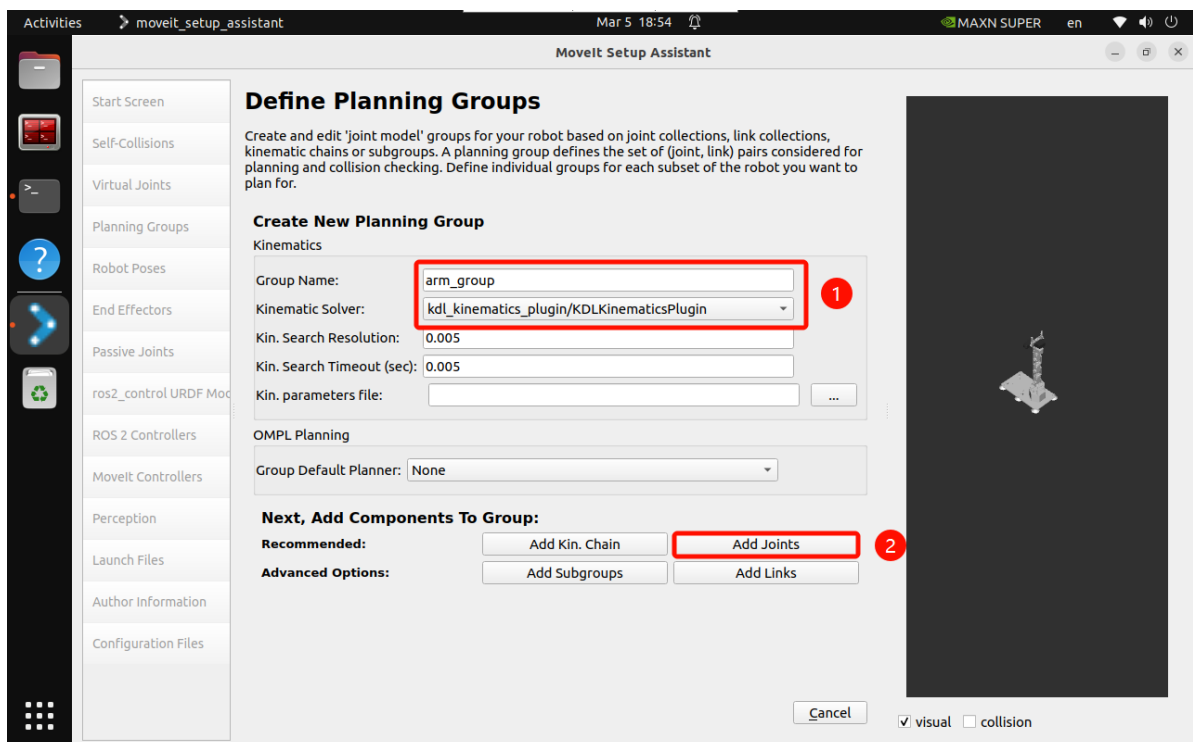
Planning groups represent a crucial step in configuring a robot's motion planning.

A planning group defines which joints and links within the robot can move together, as well as how their motion is to be planned.



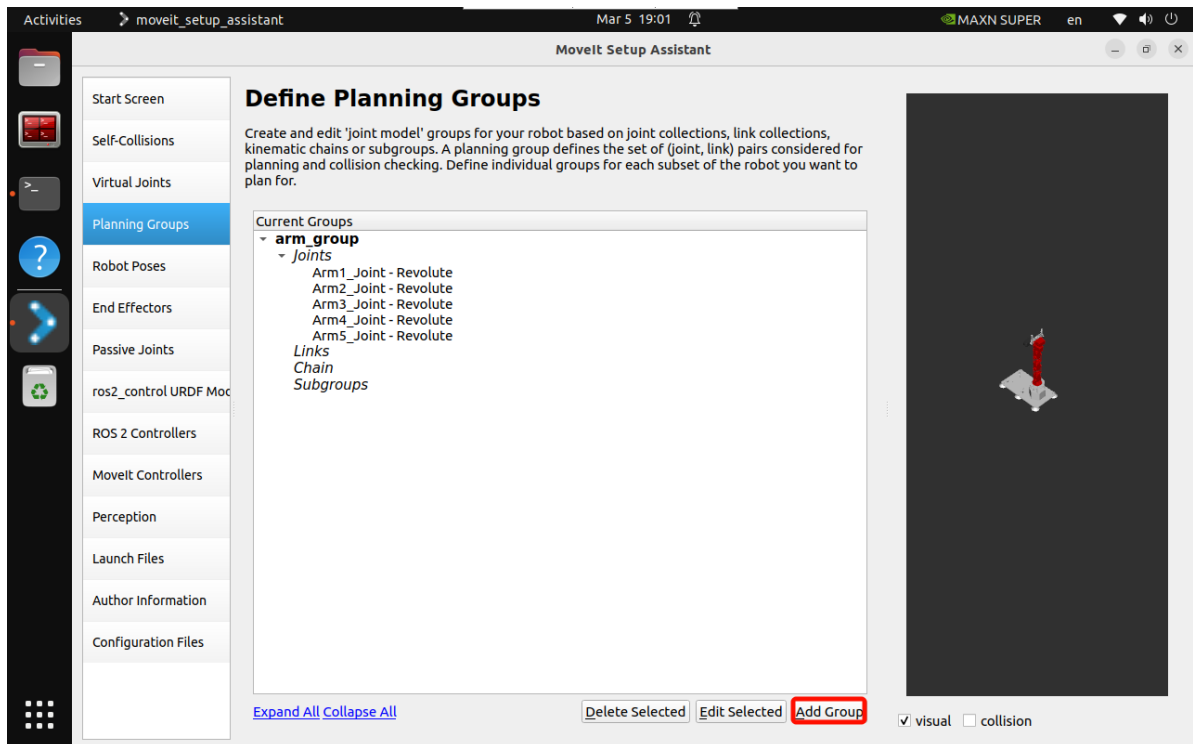
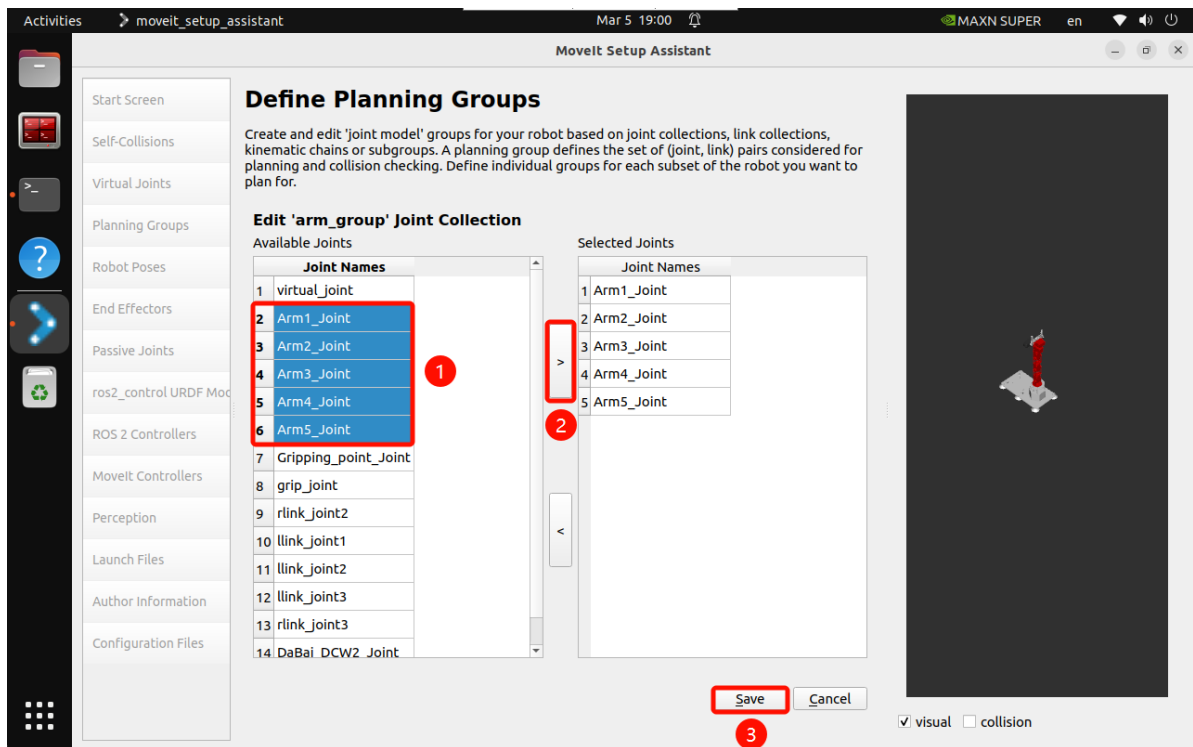
Planning Group for Robotic Arm: arm_group

- Group Name: arm_group
- Kinematic Solver: kdl_kinematics_plugin/KDLKinematicsPlugin
- Kin. Search Resolution (Sampling density in joint space): 0.005
- Kin. Search Timeout (Solution time): 0.005



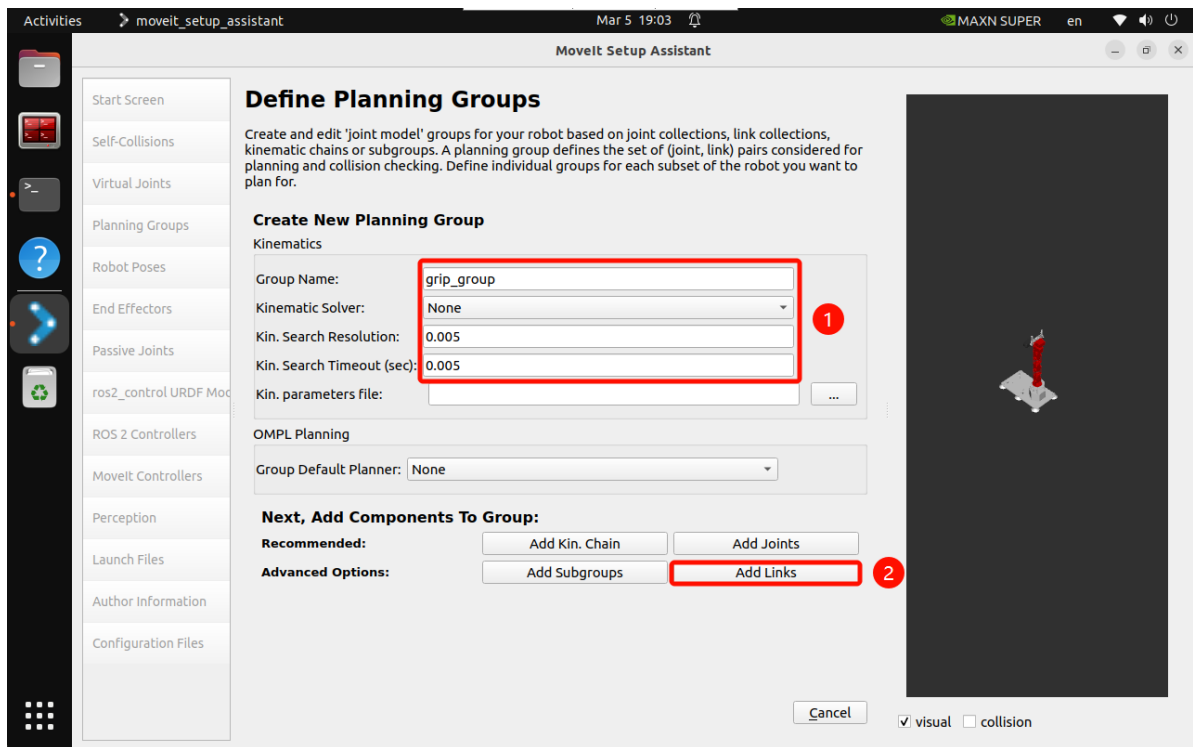
- Add Joints: Select the joints belonging to the robotic arm

Joints: Arm1_Joint, Arm2_Joint, Arm3_Joint, Arm4_Joint, Arm5_Joint



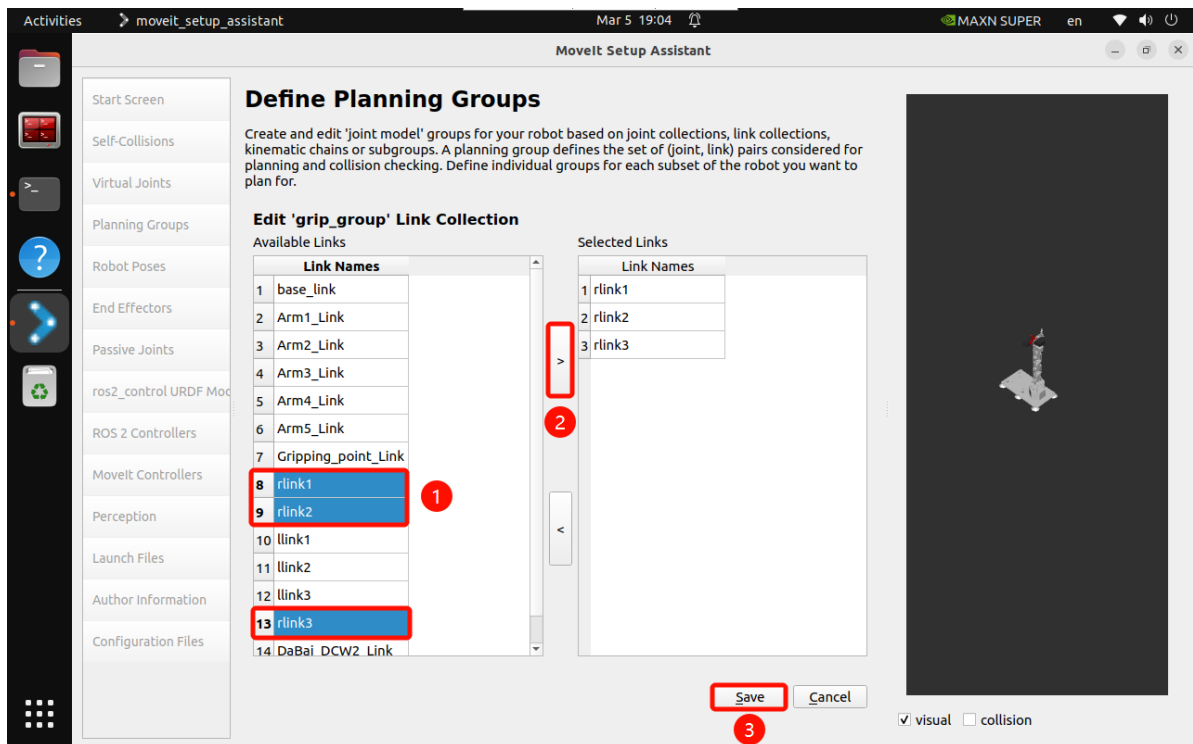
Planning Group for Gripper: grip_group

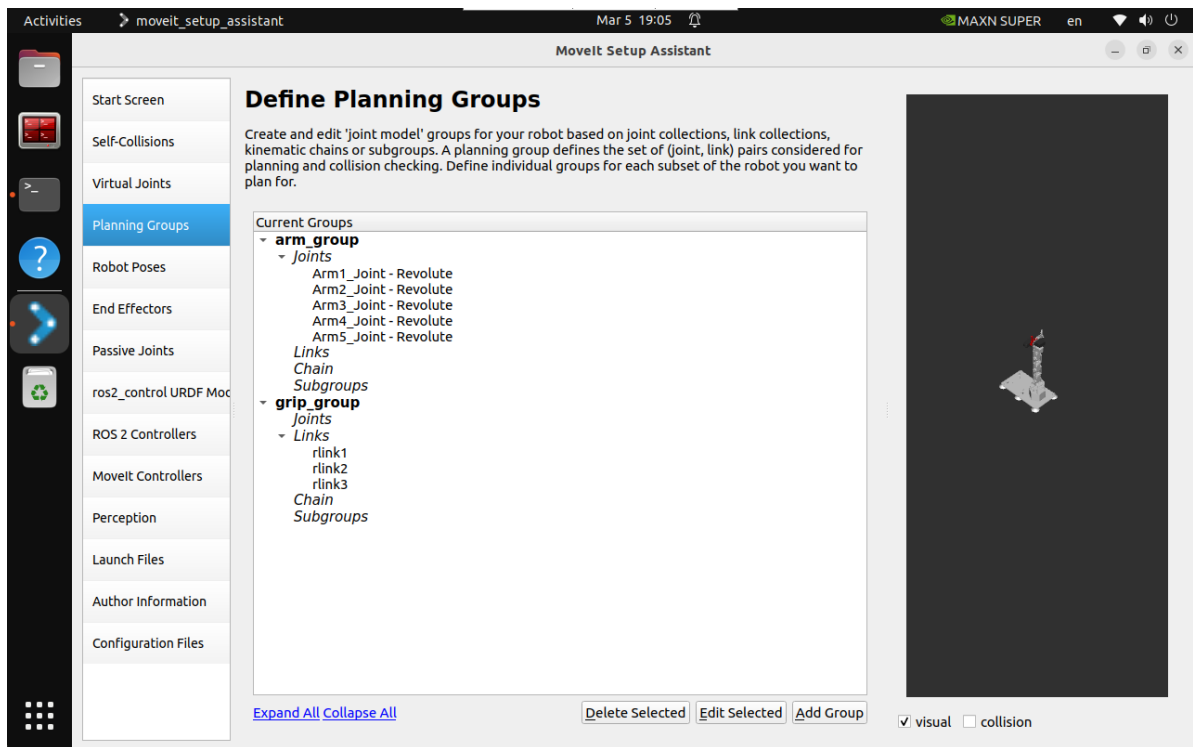
- Group Name: grip_group
- Kinematic Solver: None
- Kin. Search Resolution (Sampling density in joint space): 0.005
- Kin. Search Timeout (Solution time): 0.005



- Add Links: Select the gripper's connecting rods.

Links: rlink1, rlink2, rlink3

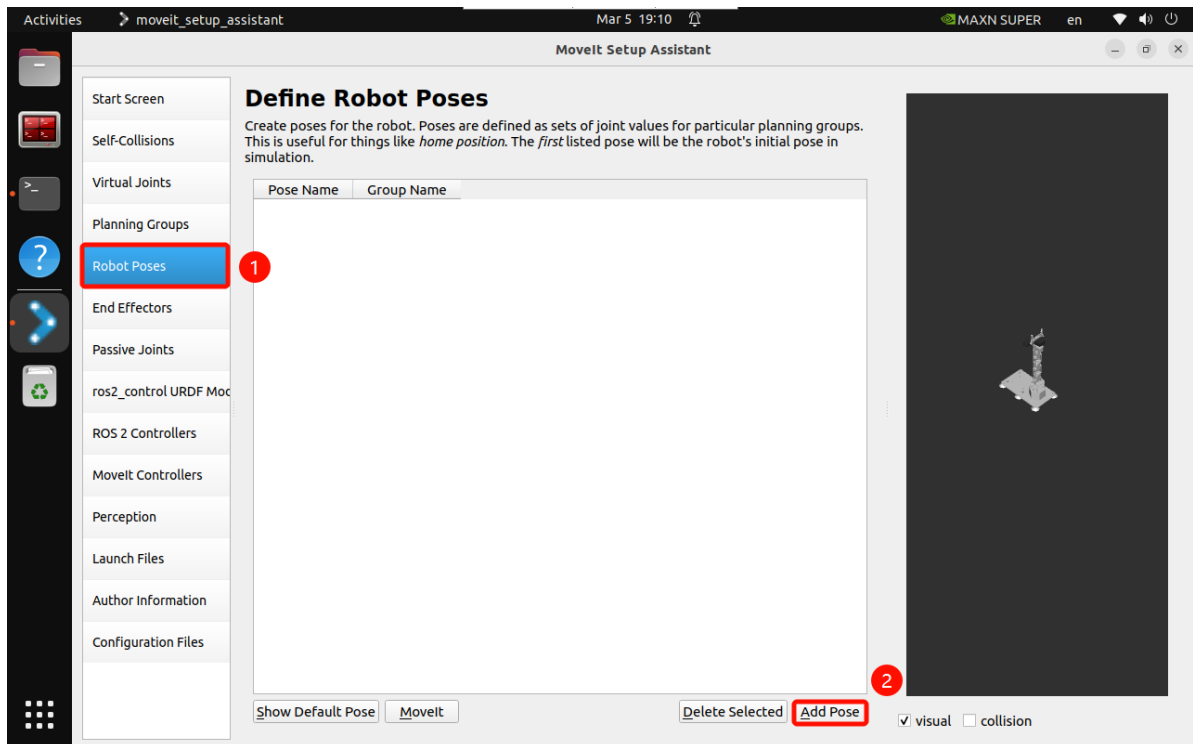




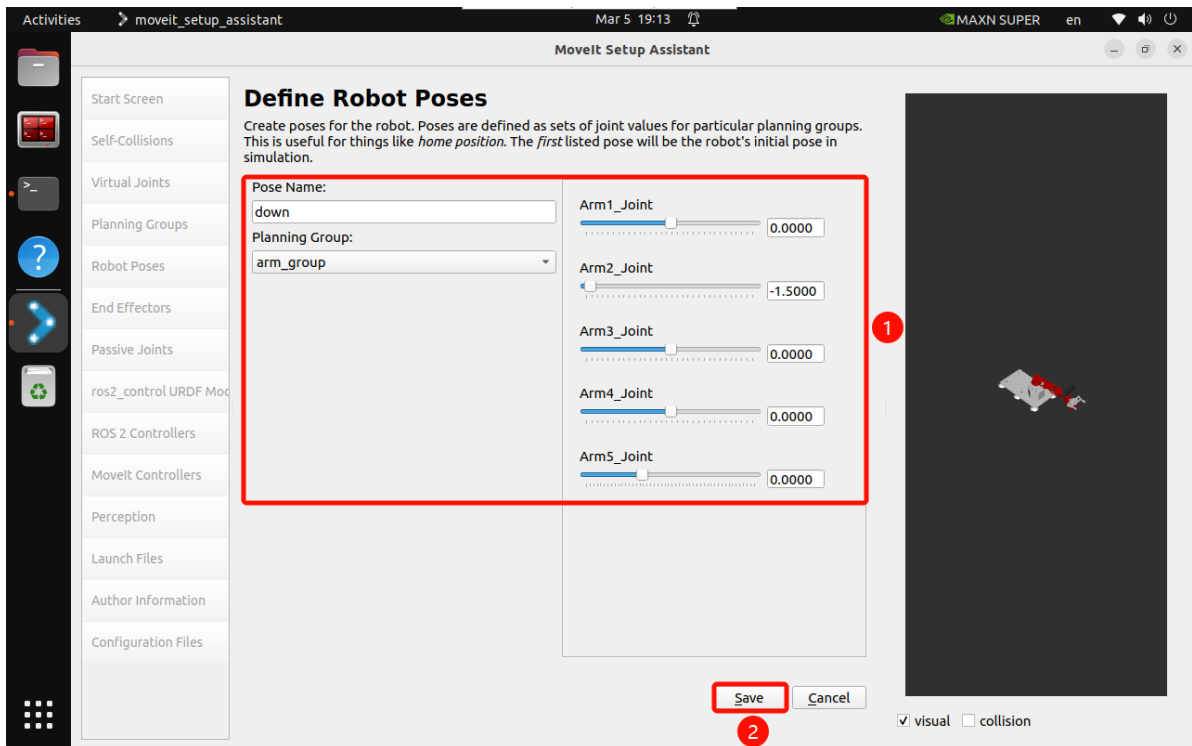
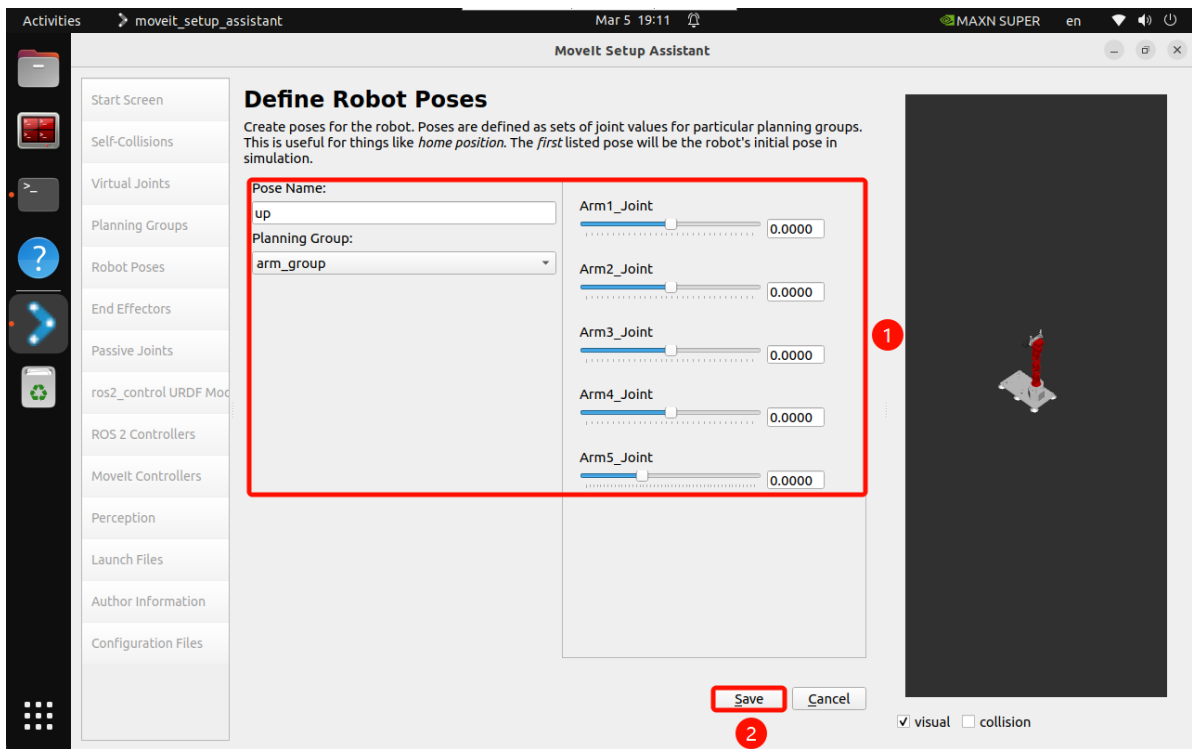
2.5 Robot Positions

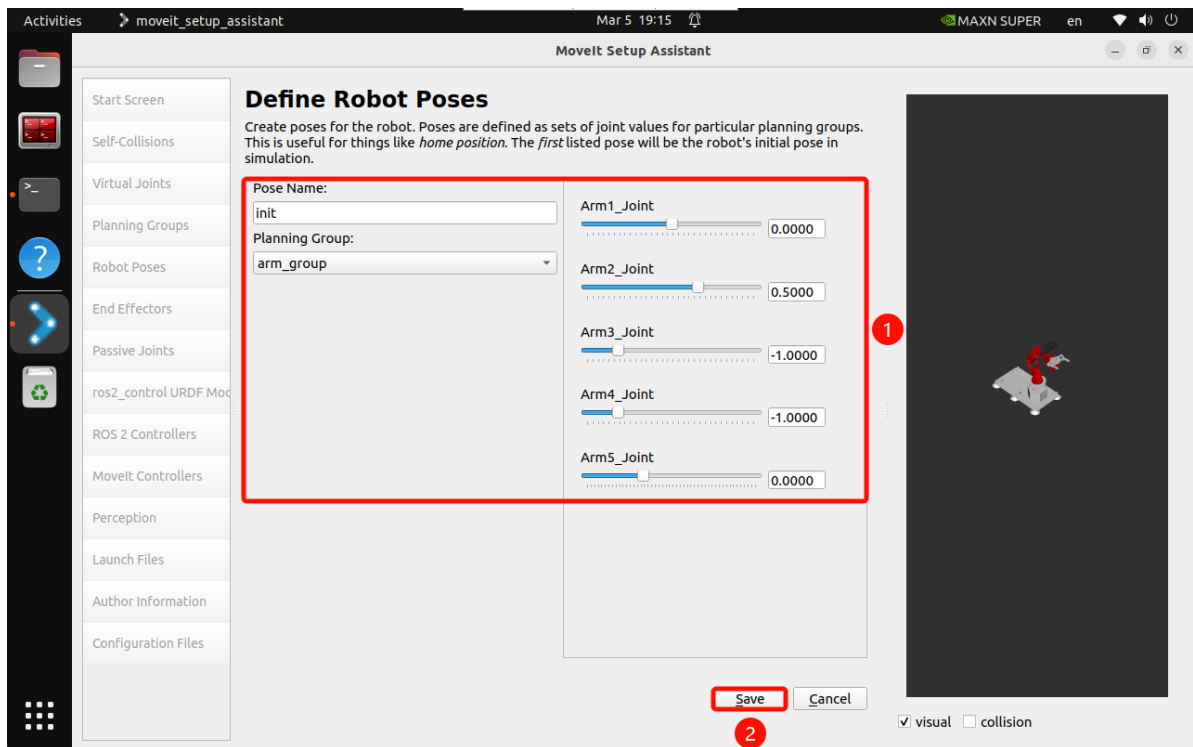
Robot Arm Positions

Configure three sets of predefined robot arm poses within the robot configuration settings:



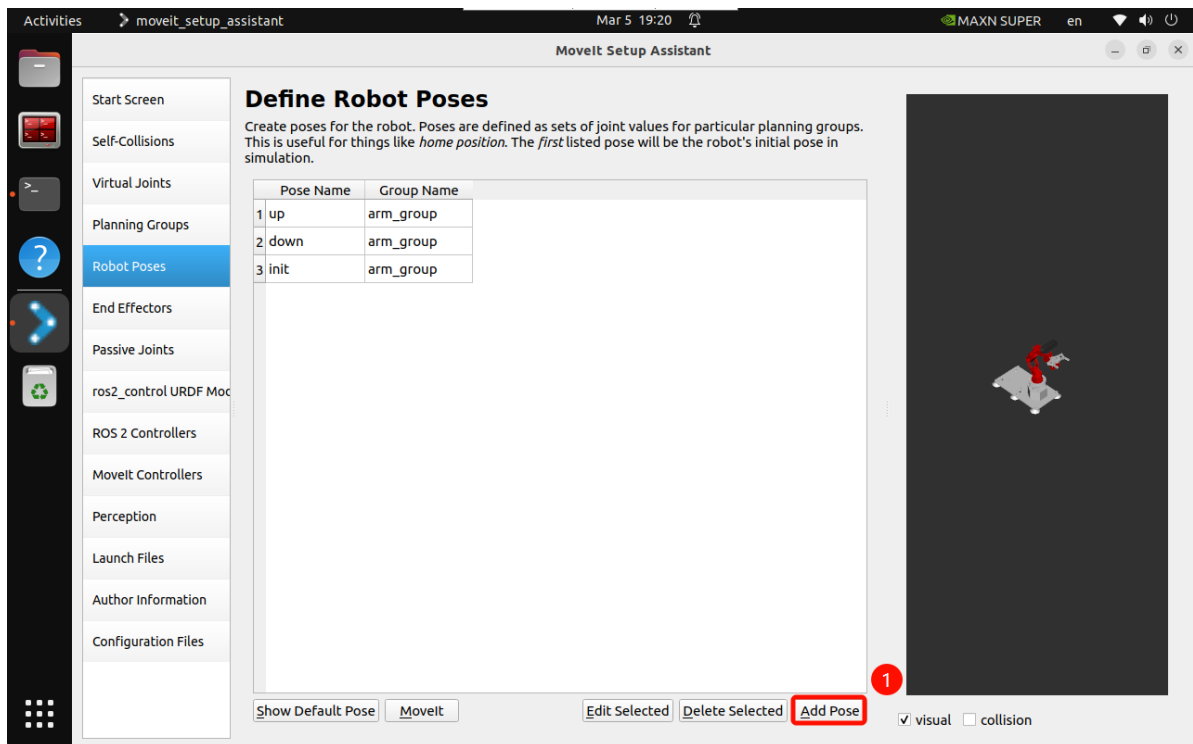
You can manually adjust the sliders to set the state of each individual joint: Up, Down, or Init.



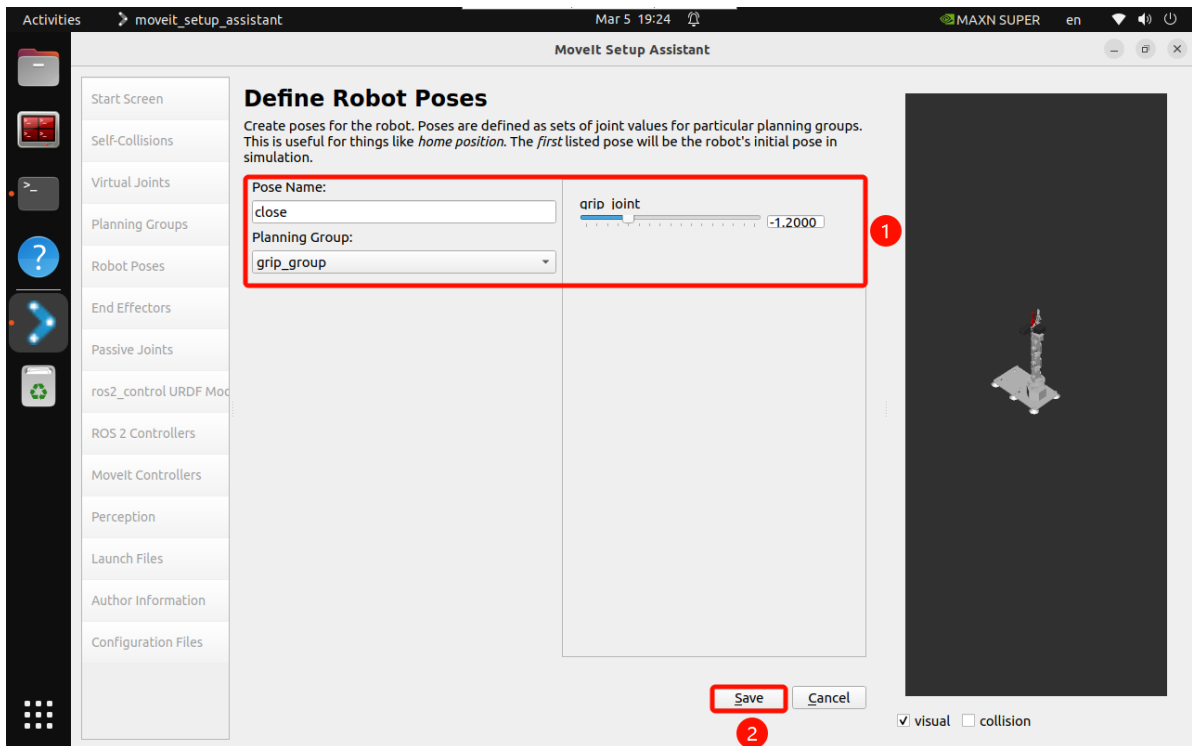
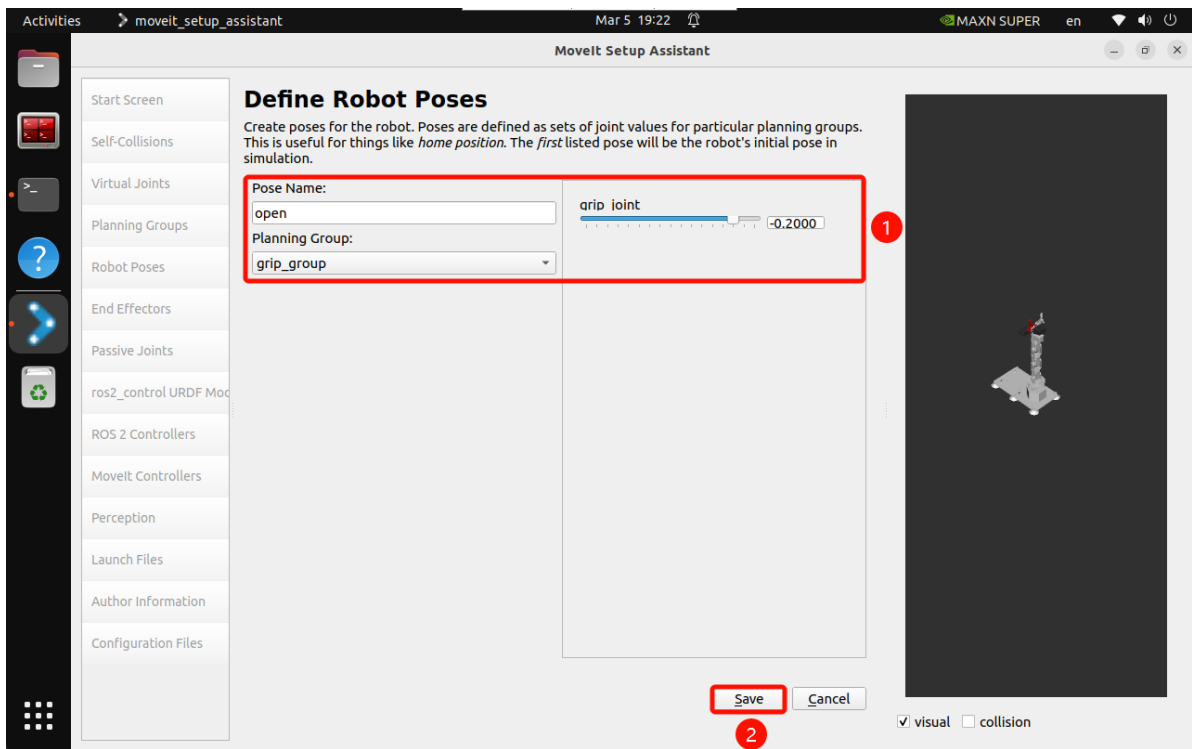


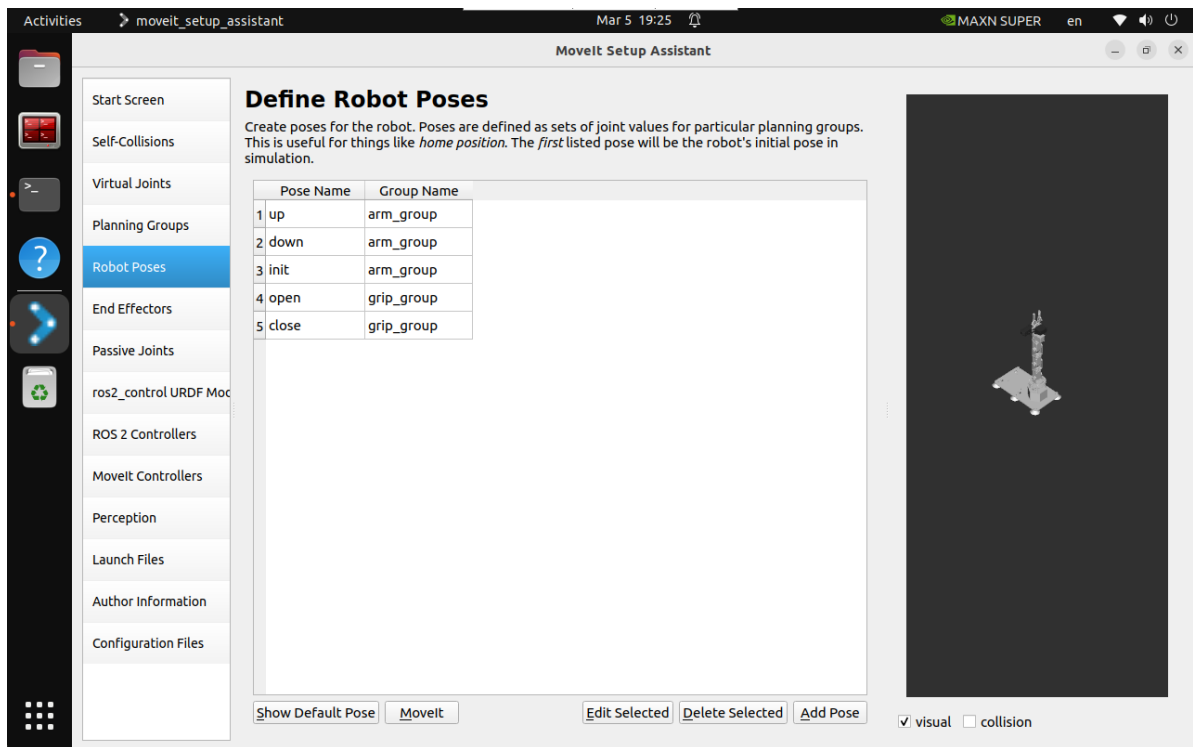
Gripper Positions

Configure two sets of predefined gripper poses within the robot configuration settings:



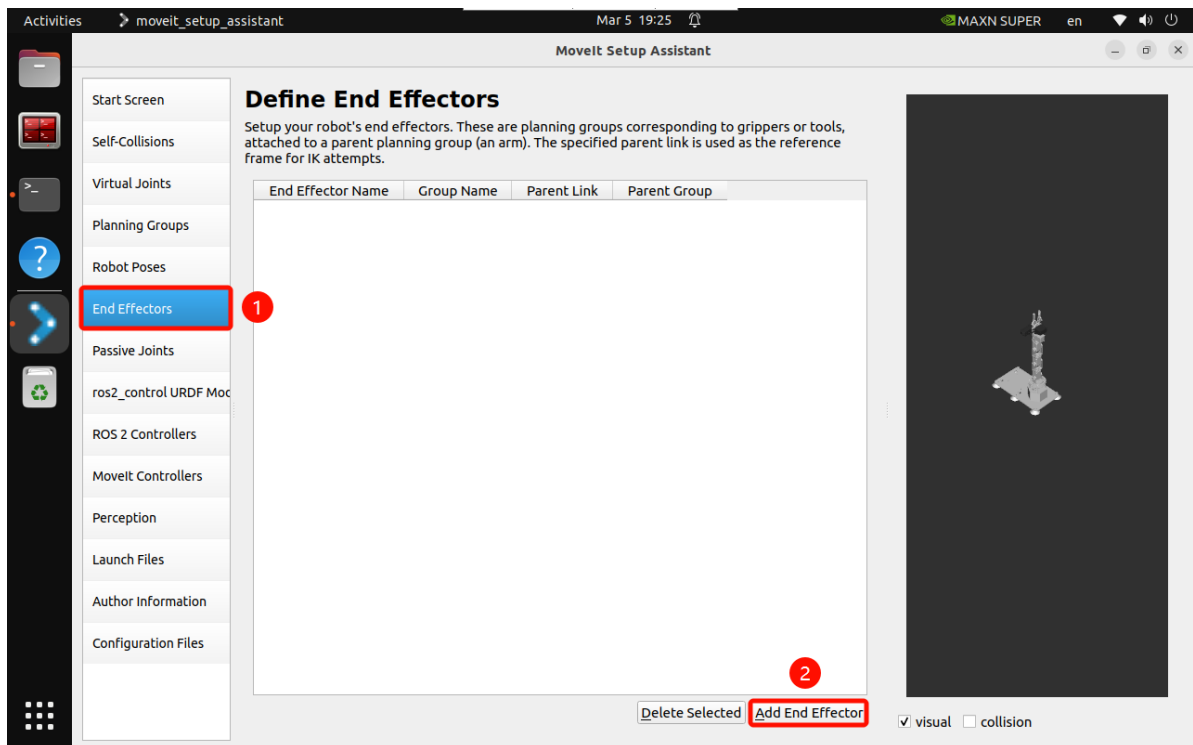
You can manually adjust the sliders to set the gripper state: Open or Close.





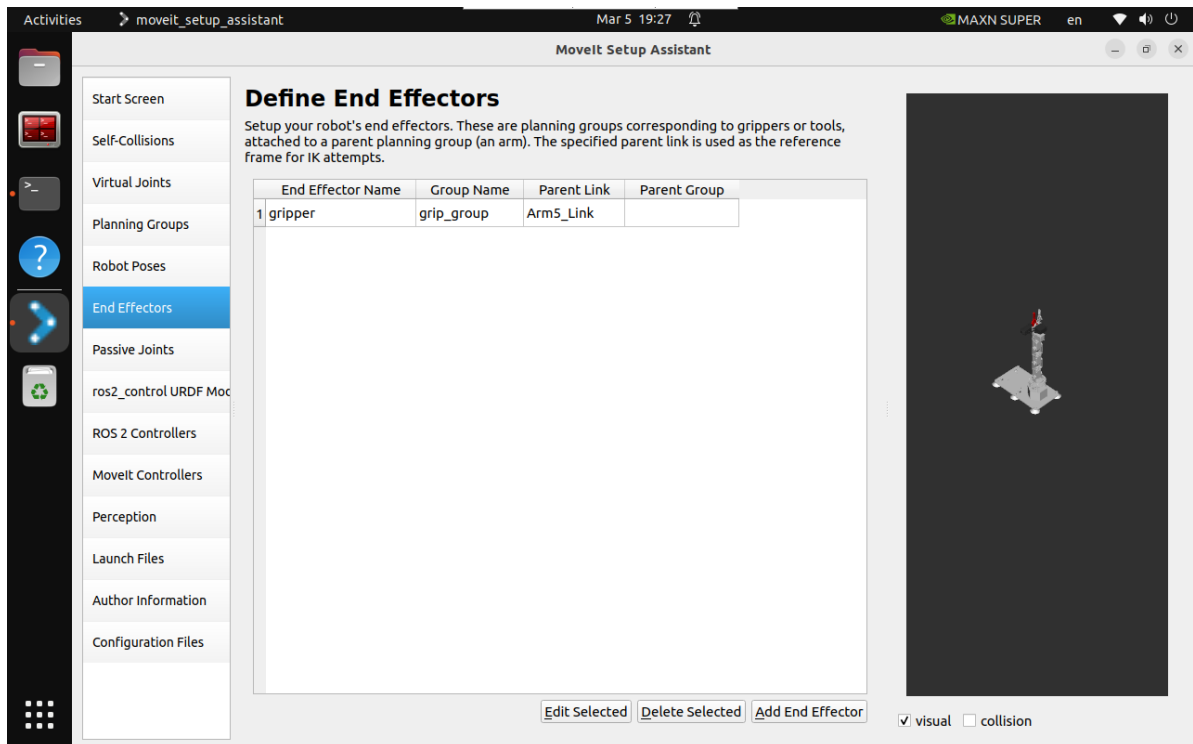
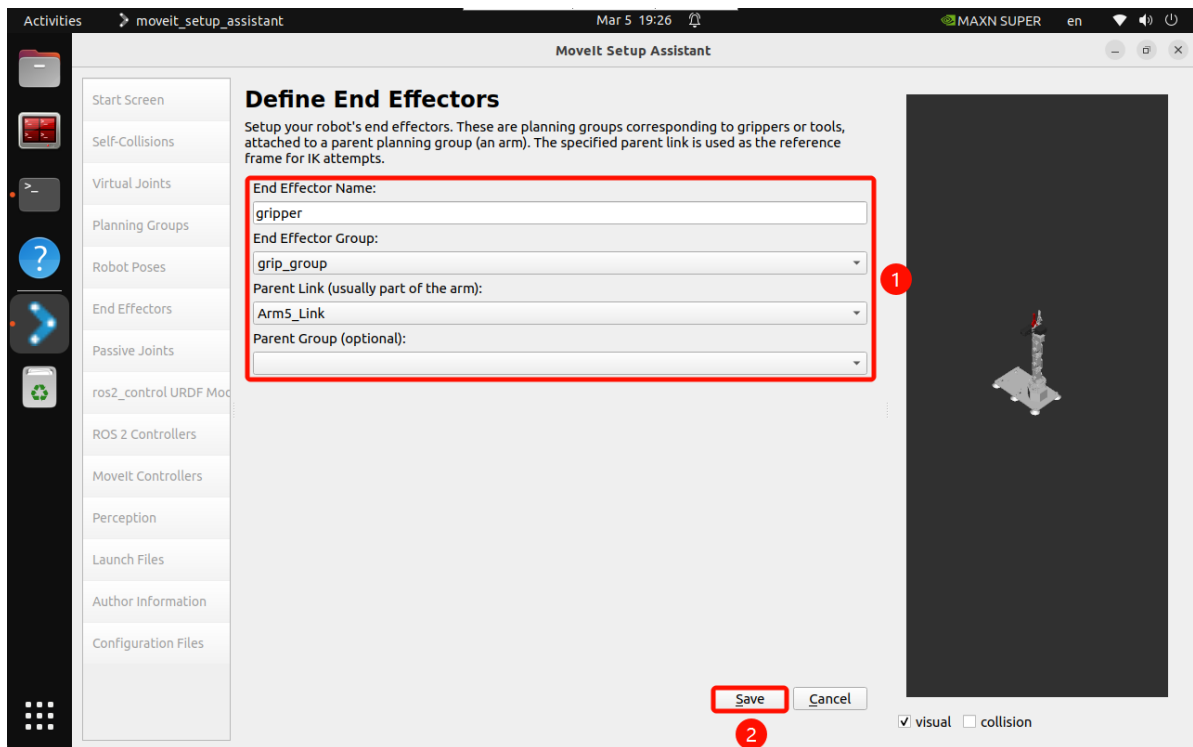
2.6. End Effector

Set the gripper group as the end effector:



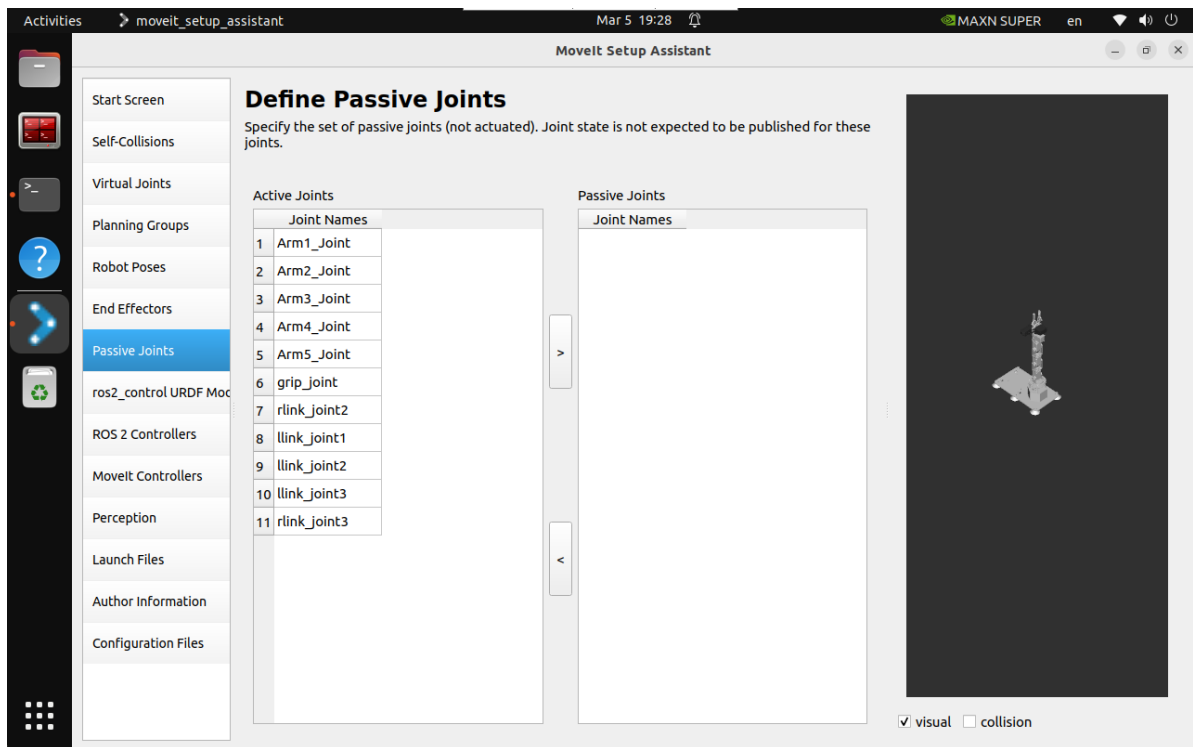
End Effector: gripper

- End Effector Name: gripper
- End Effector Group: gripp_group
- Parent Link: Arm5_Link
- Parent Group: None



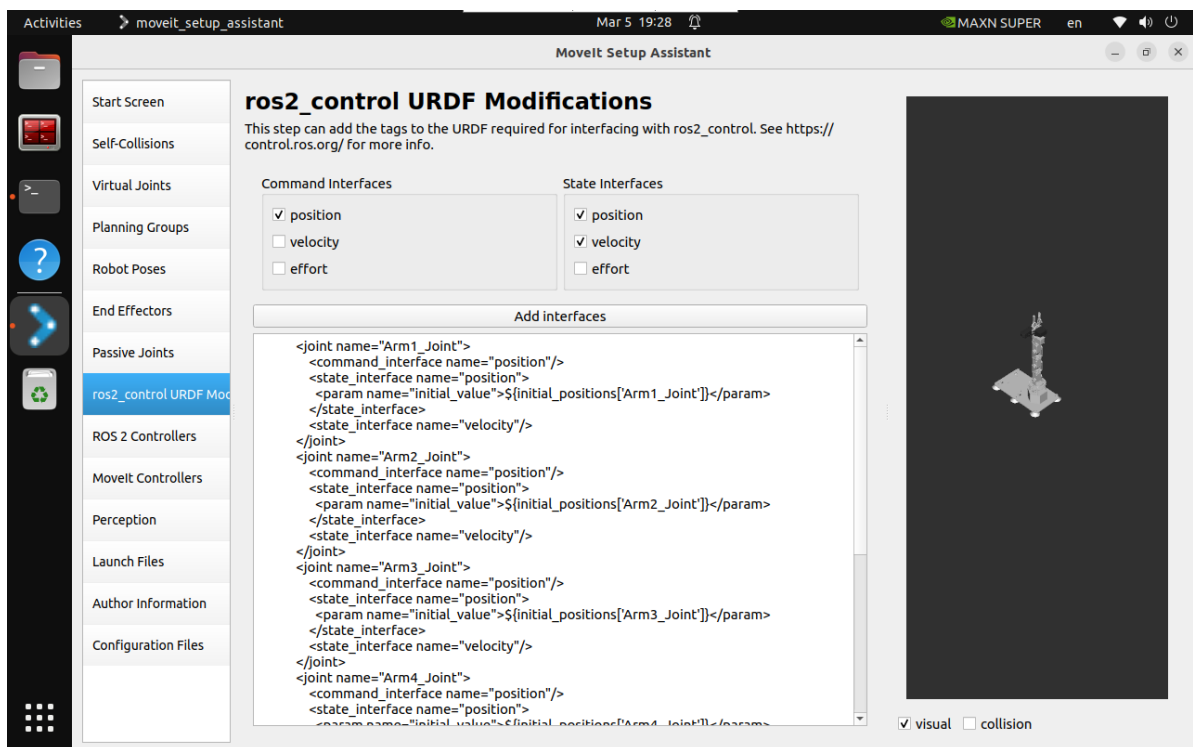
2.7. Passive Joints

Passive joints are non-actuated joints that cannot be directly controlled. If the robotic arm has no passive joints, simply skip this step.



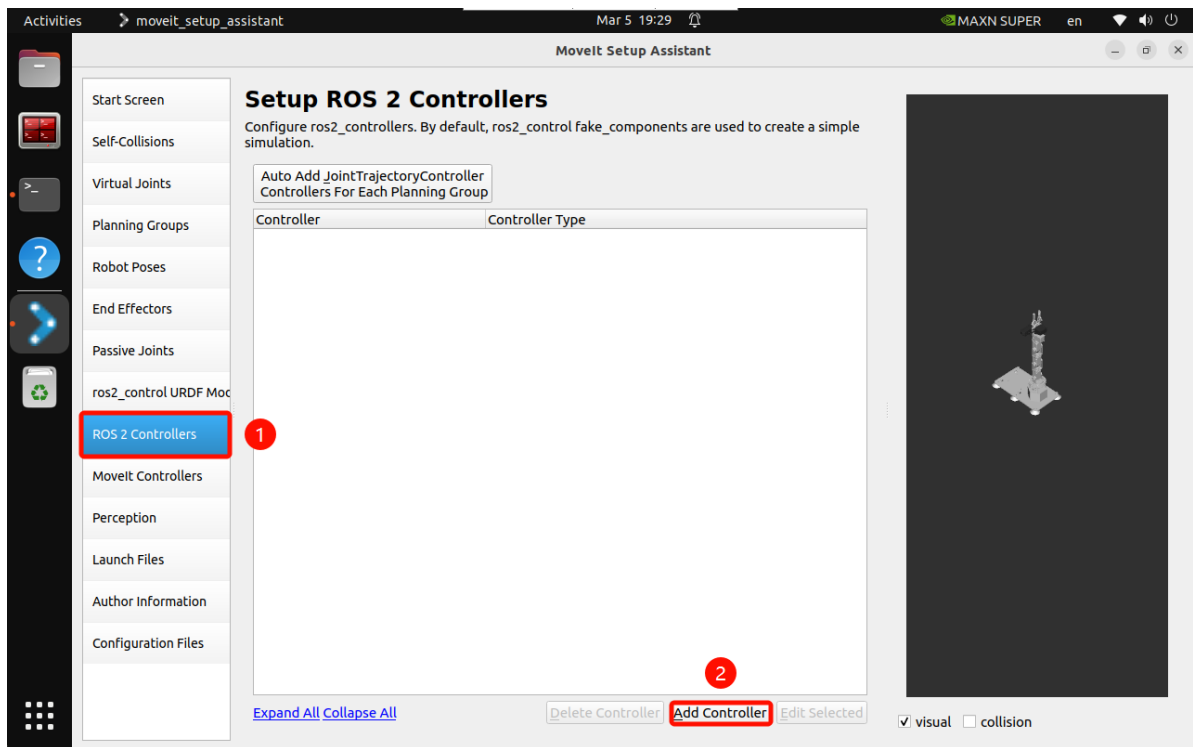
2.8. URDF File

The MoveIt Setup Assistant automatically configures the command interfaces and state interfaces for each joint; use the default generated information.



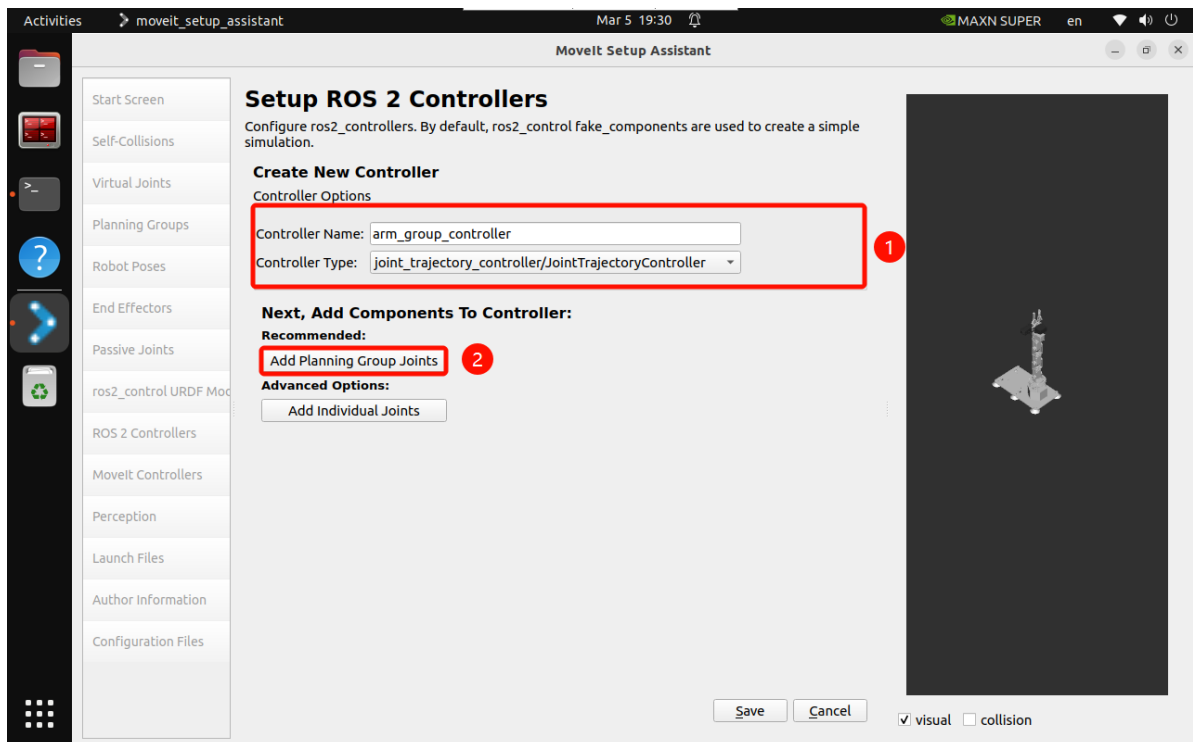
2.9. ROS2 Controllers

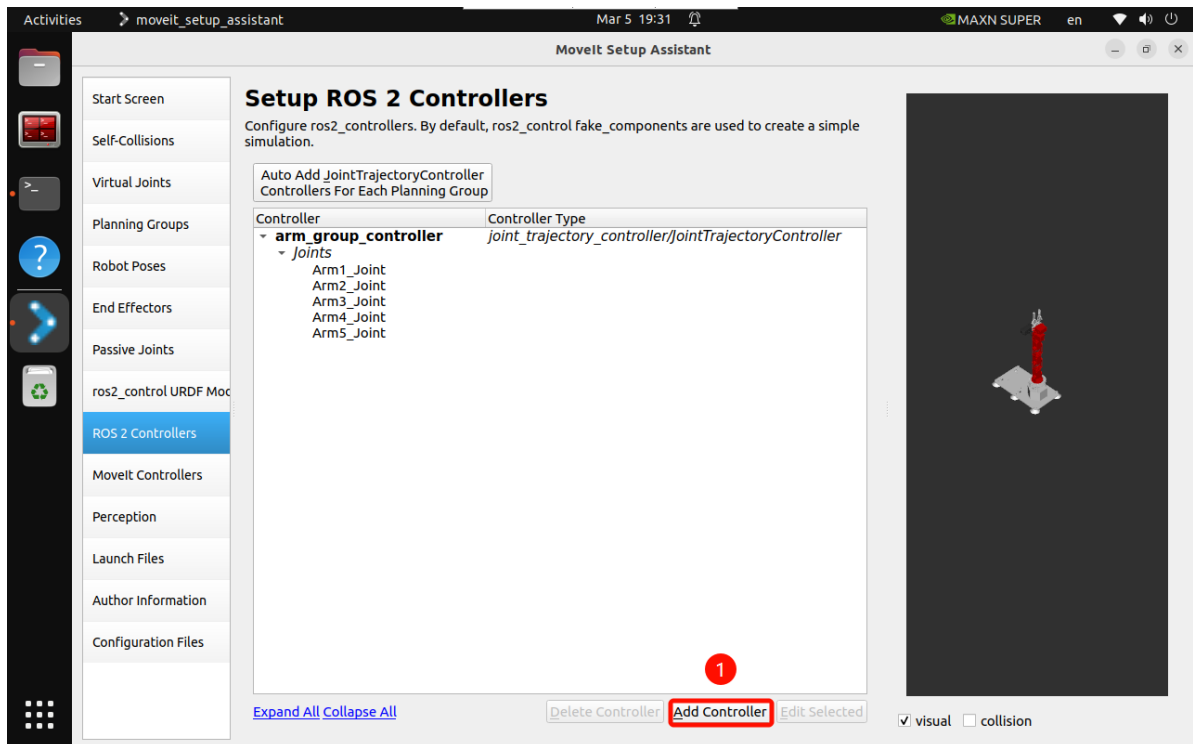
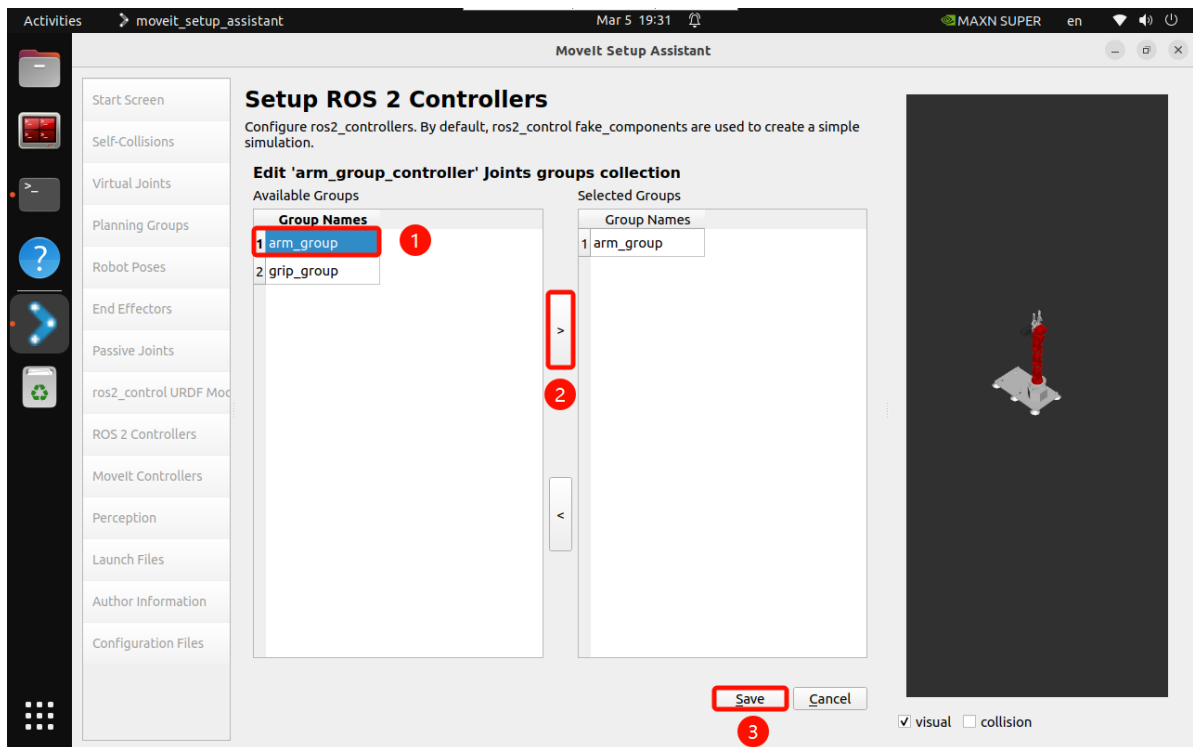
ROS2 Controllers is a framework designed for real-time robot control; it can be used to automatically generate simulation controllers to drive the robot's joints.



Add Arm Controller:

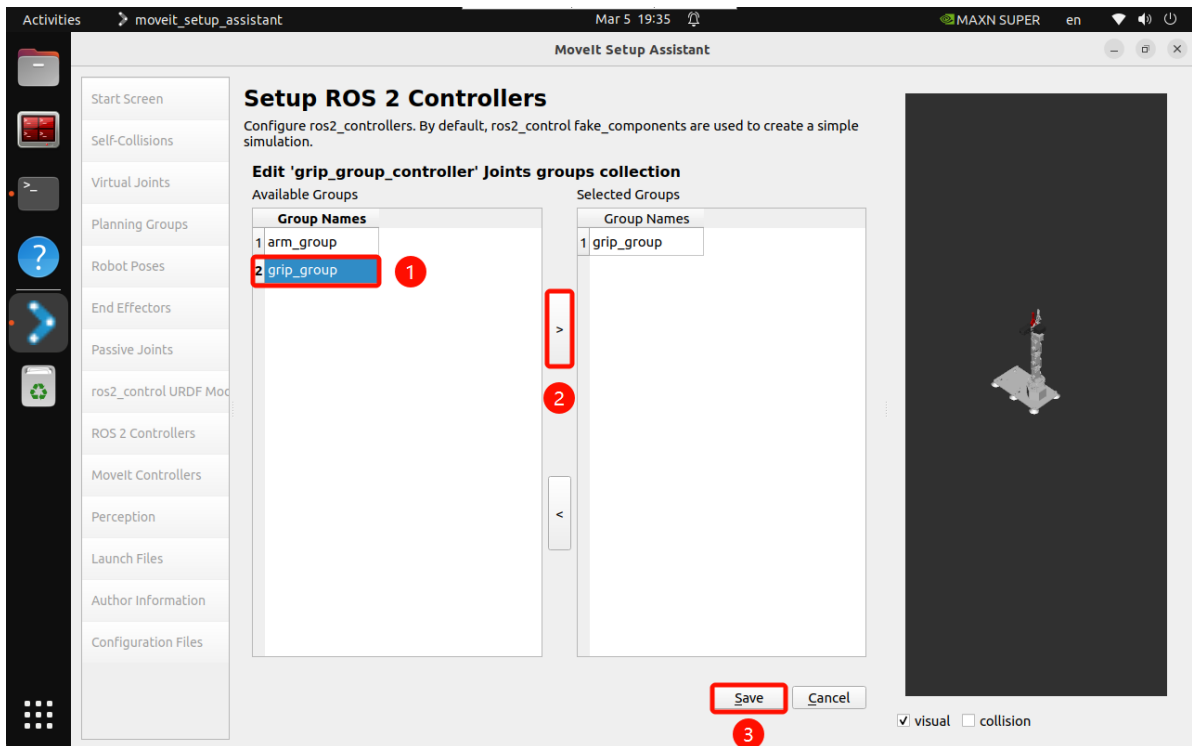
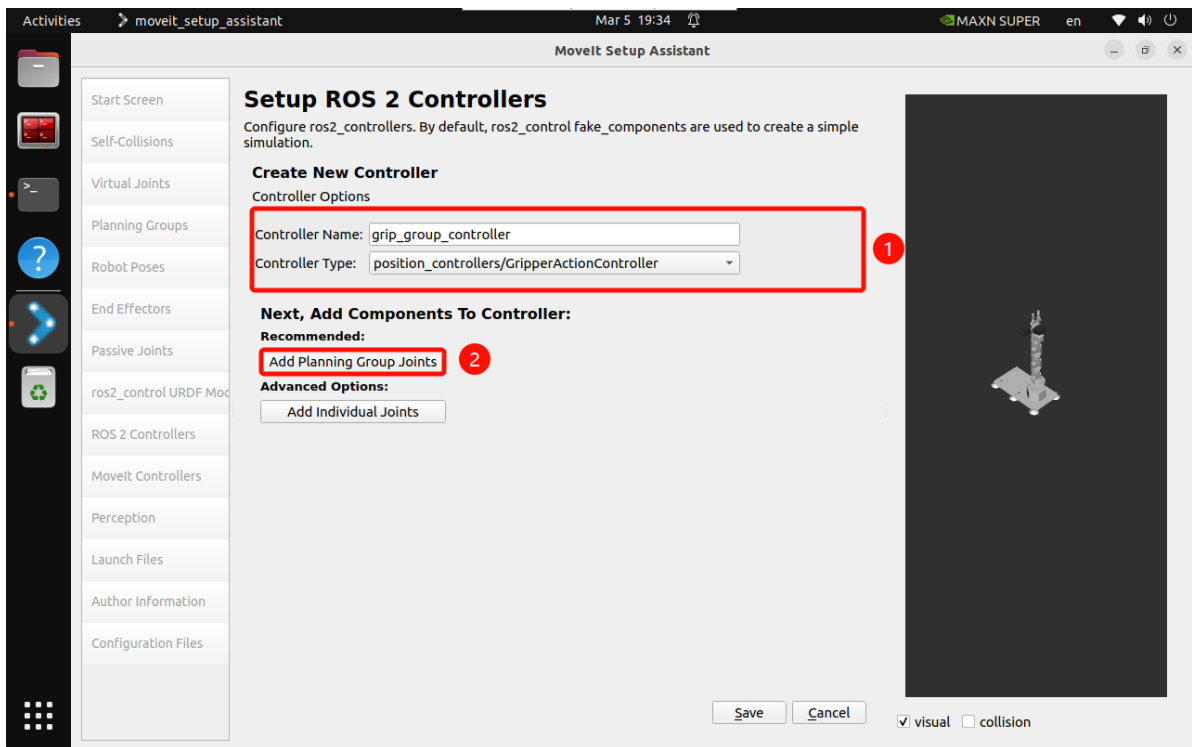
- Controller Name: arm_group_controller
- Controller Type: joint_trajectory_controller/JointTrajectoryController
- Add Planning Group Joints: arm_group

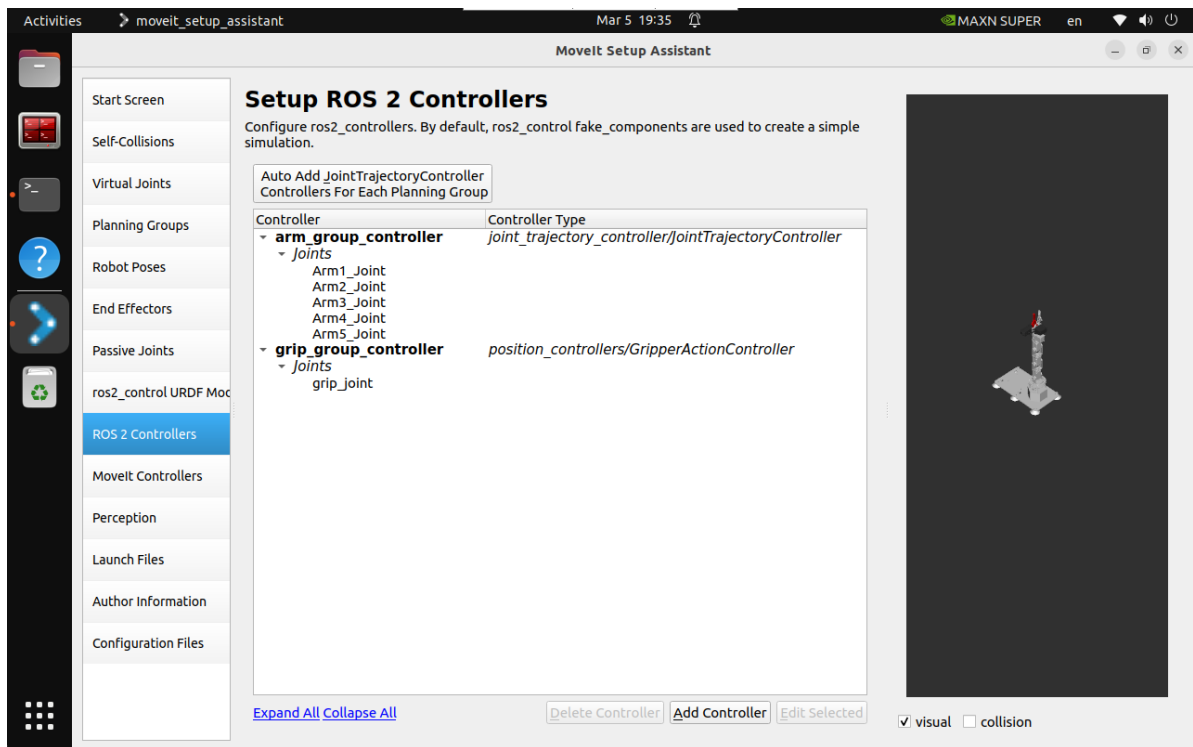




Add Gripper Controller:

- Controller Name: gripper_group_controller
- Controller Type: position_controllers/GripperActionController
- Add Planning Group Joints: gripper_group

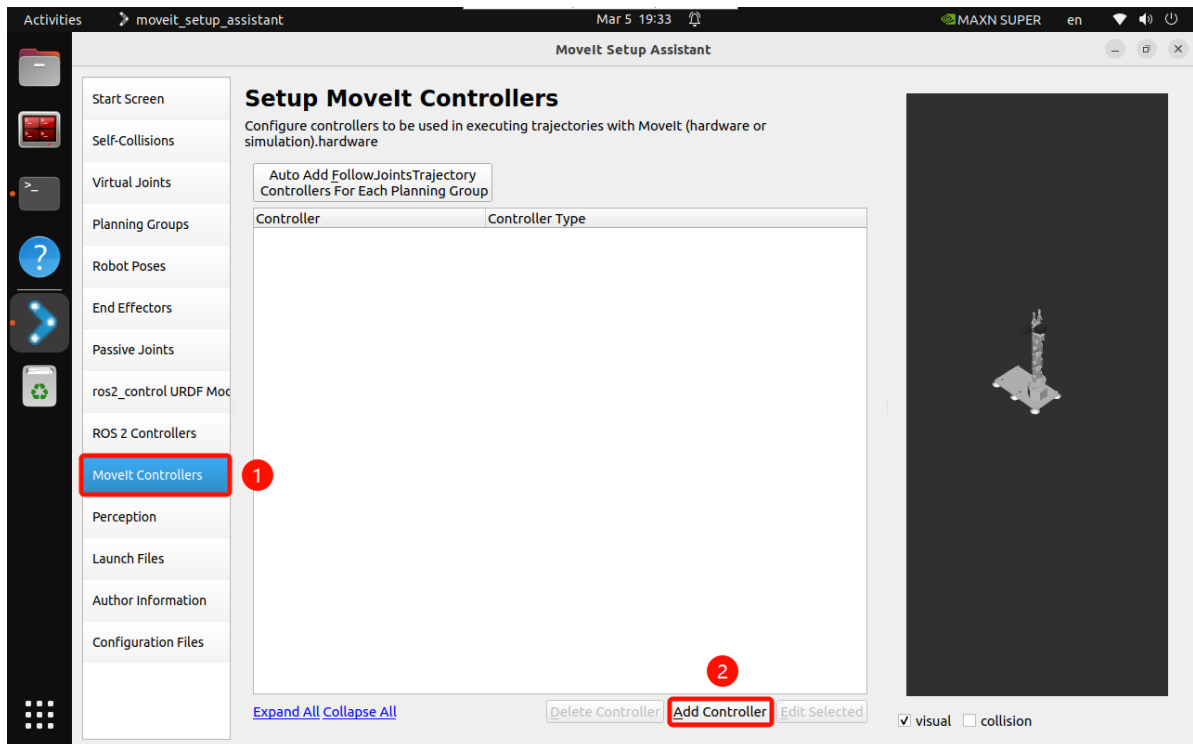




2.10: MoveIt Controllers

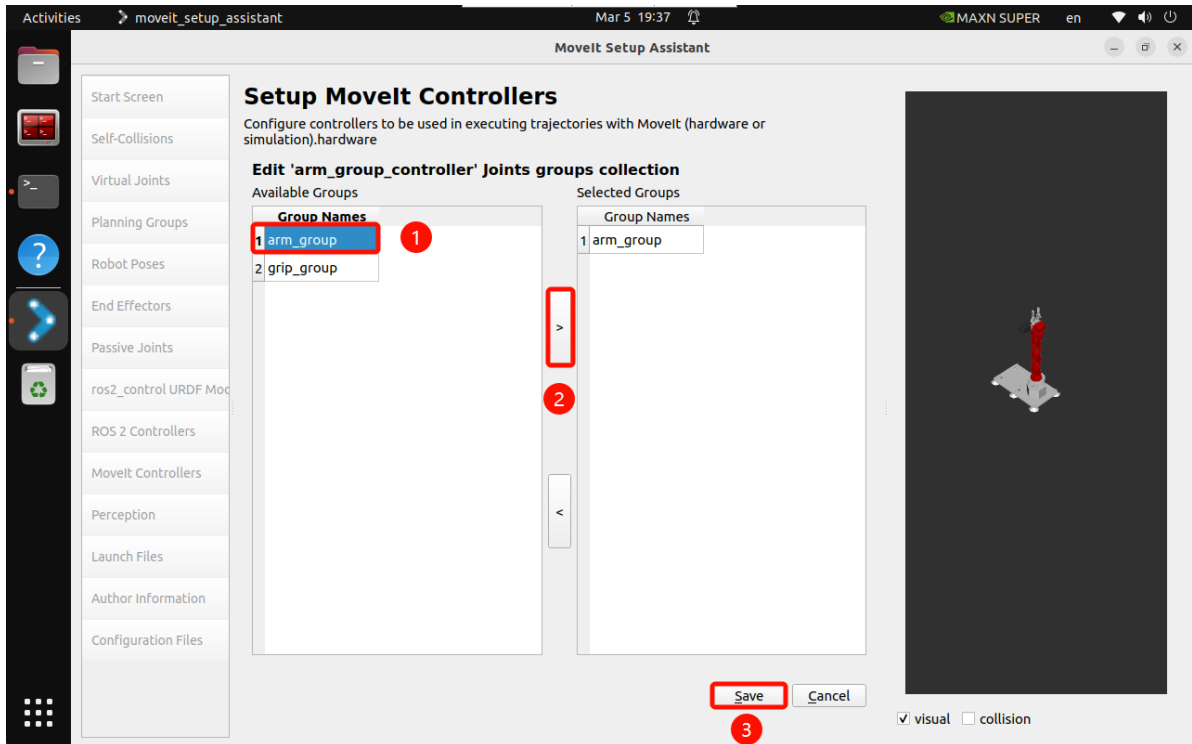
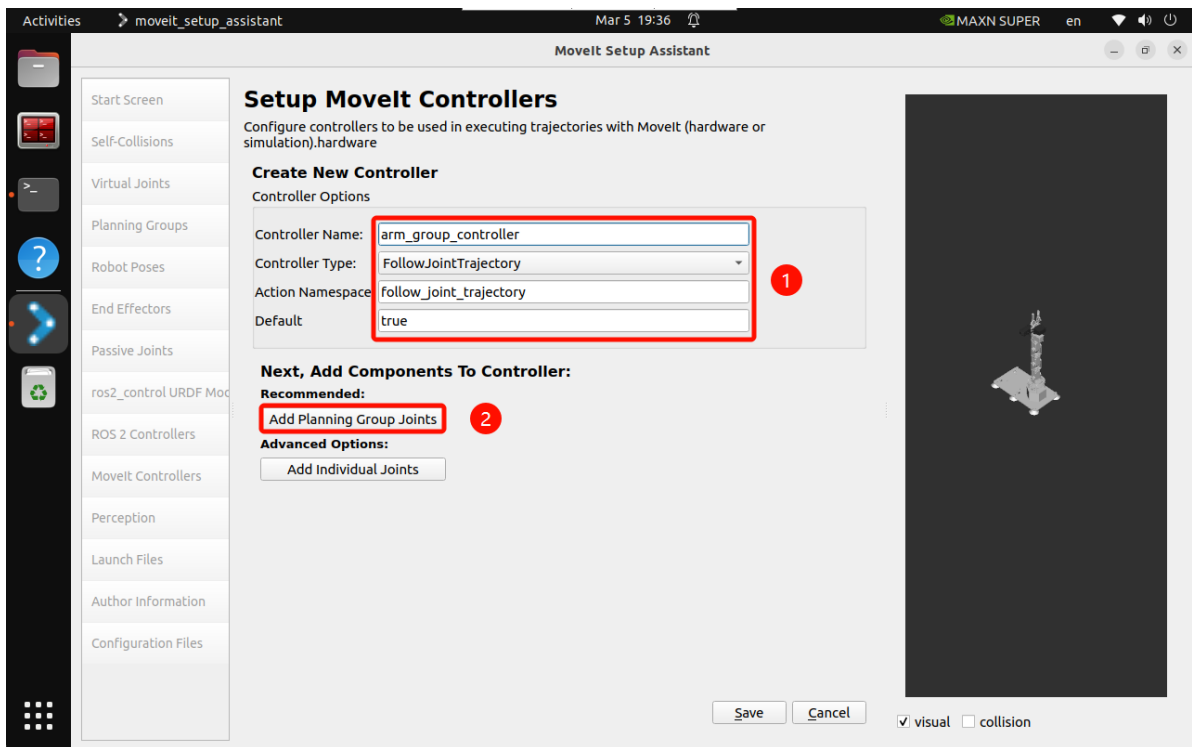
MoveIt requires a trajectory controller featuring the `FollowJointTrajectoryAction` interface to execute planned trajectories; this interface transmits the generated trajectories to the robot's ROS 2 controllers.

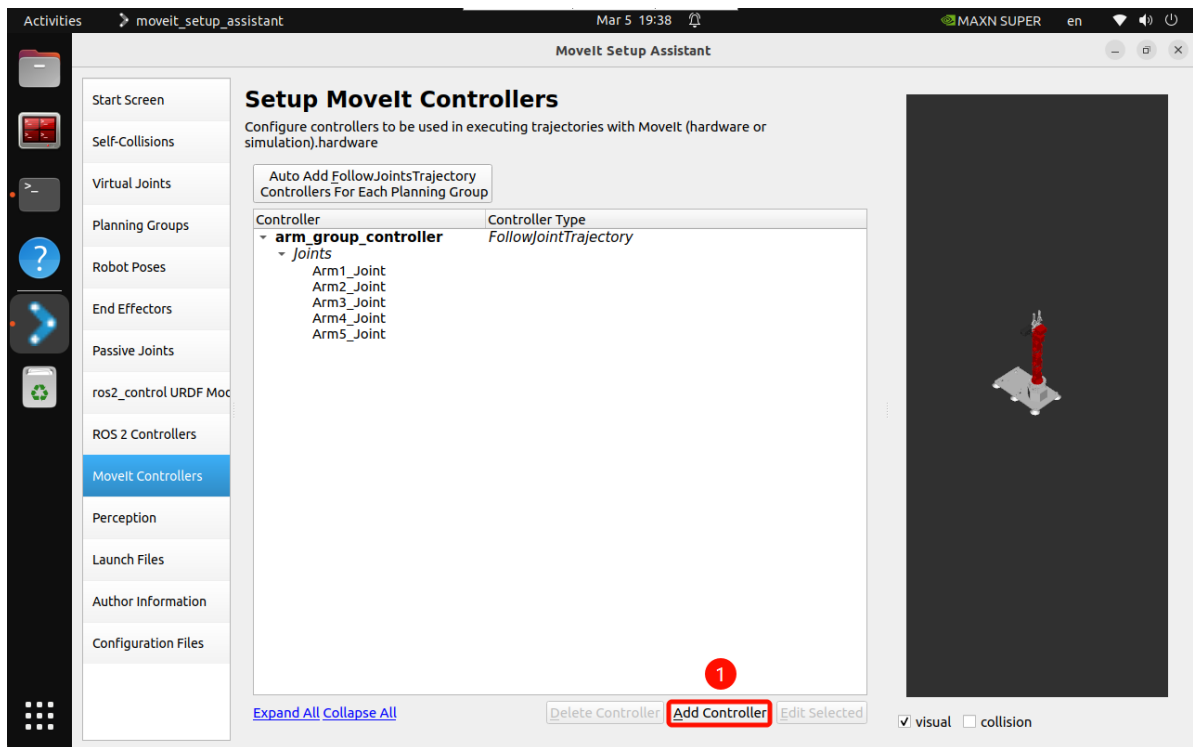
When adding MoveIt Controllers, ensure that the controller names match the names configured within the ROS 2 Controllers.



Add Arm Controller:

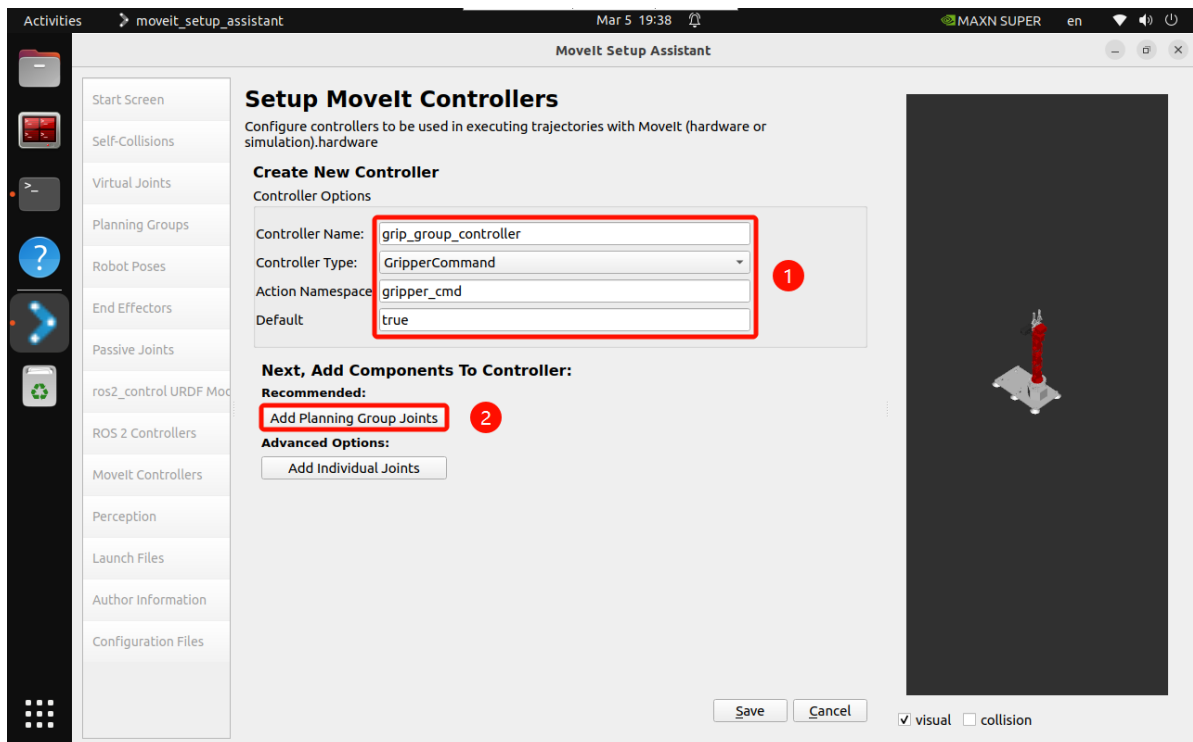
- Controller Name: arm_group_controller
- Controller Type: FollowJointTrajectory
- Add Planning Group Joints: arm_group

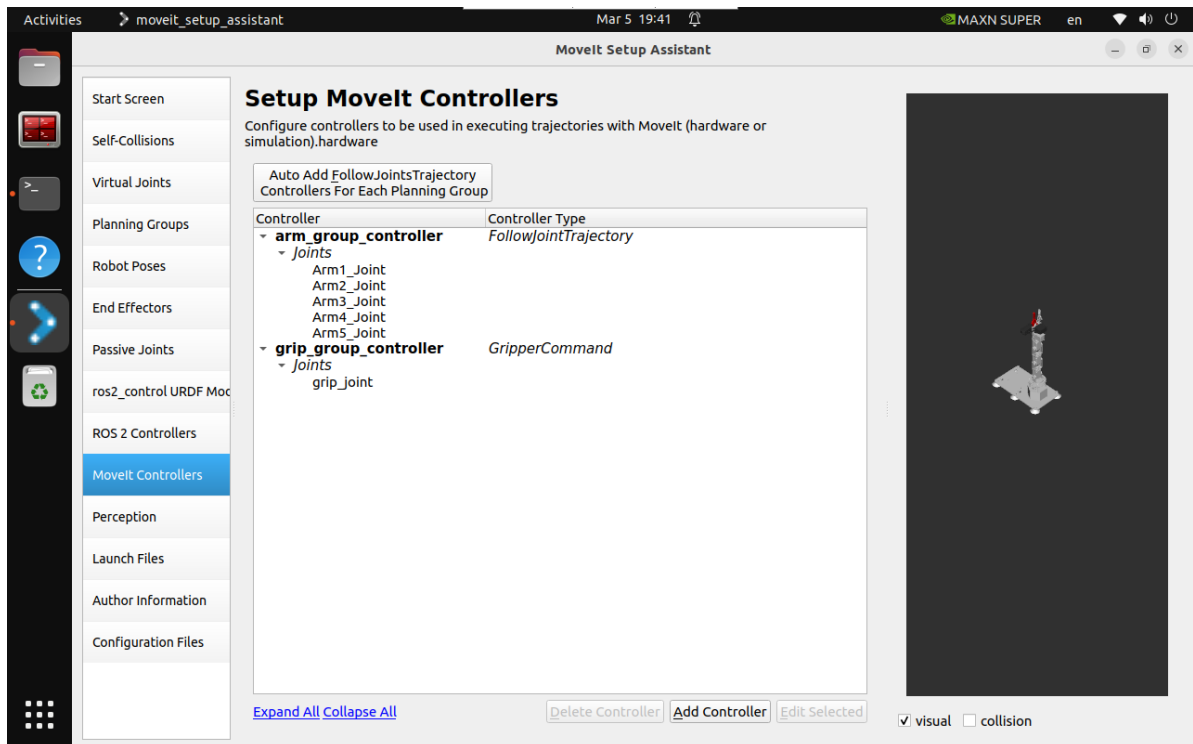
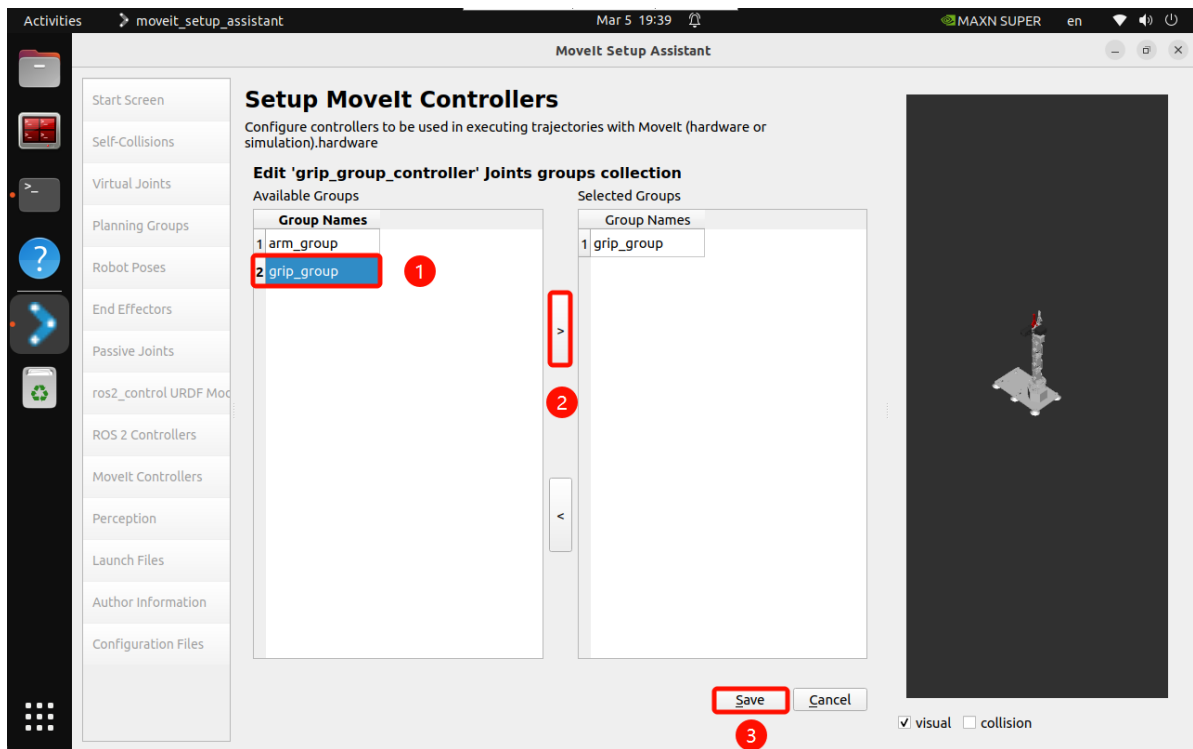




Add Gripper Controller:

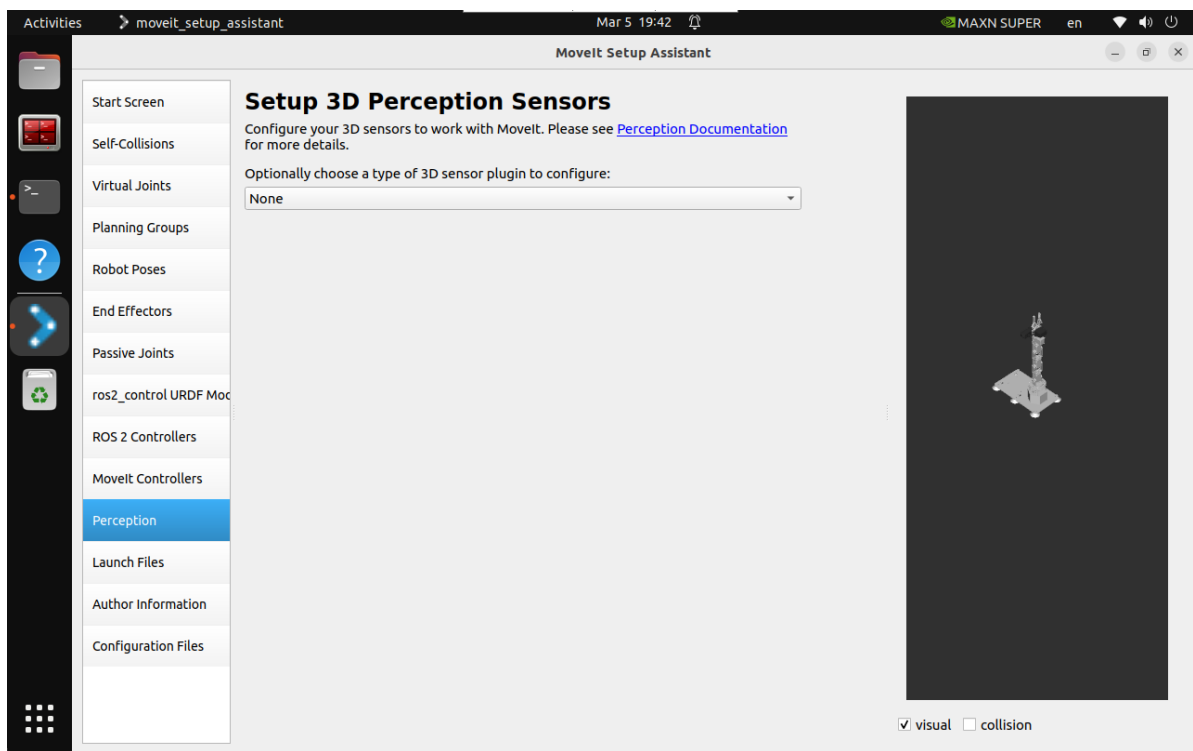
- Controller Name: grip_group_controller
- Controller Type: Gripper Command
- Add Planning Group Joints: grip_group





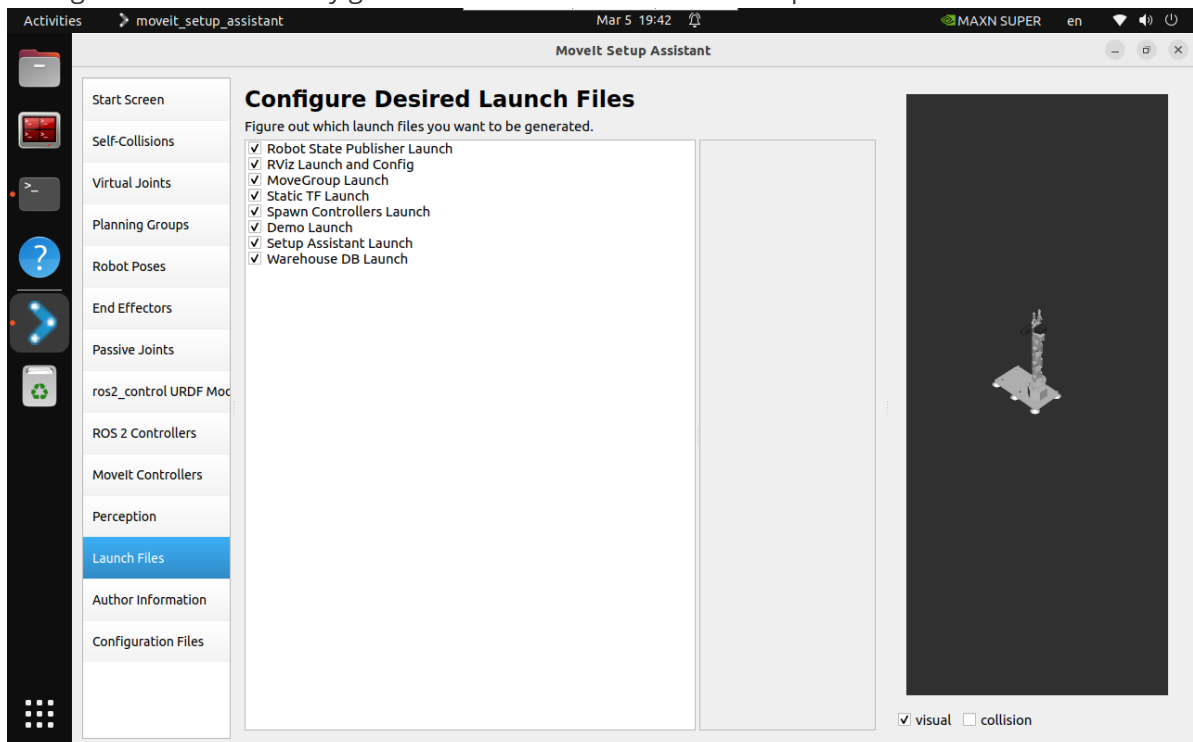
2.11 Sensors

Configure settings for the 3D sensors used by the robot: No configuration required.



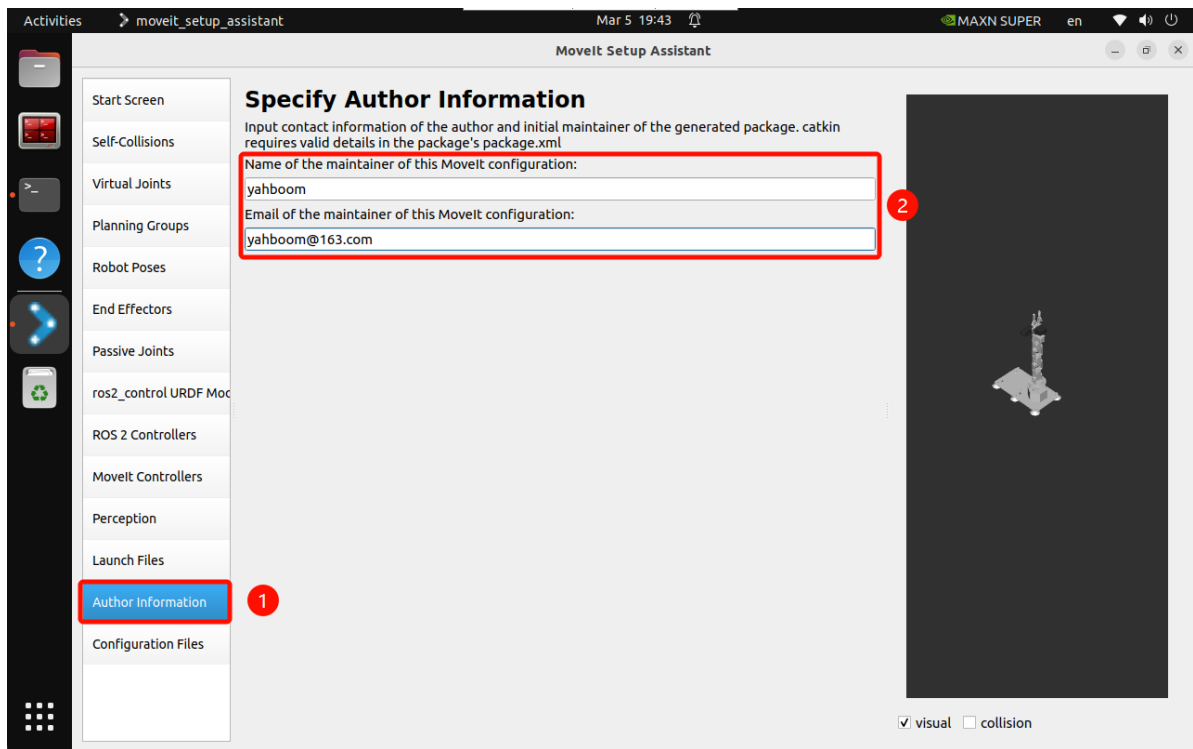
2.12 Launch Files

Configure the automatically generated launch files: Use default options.



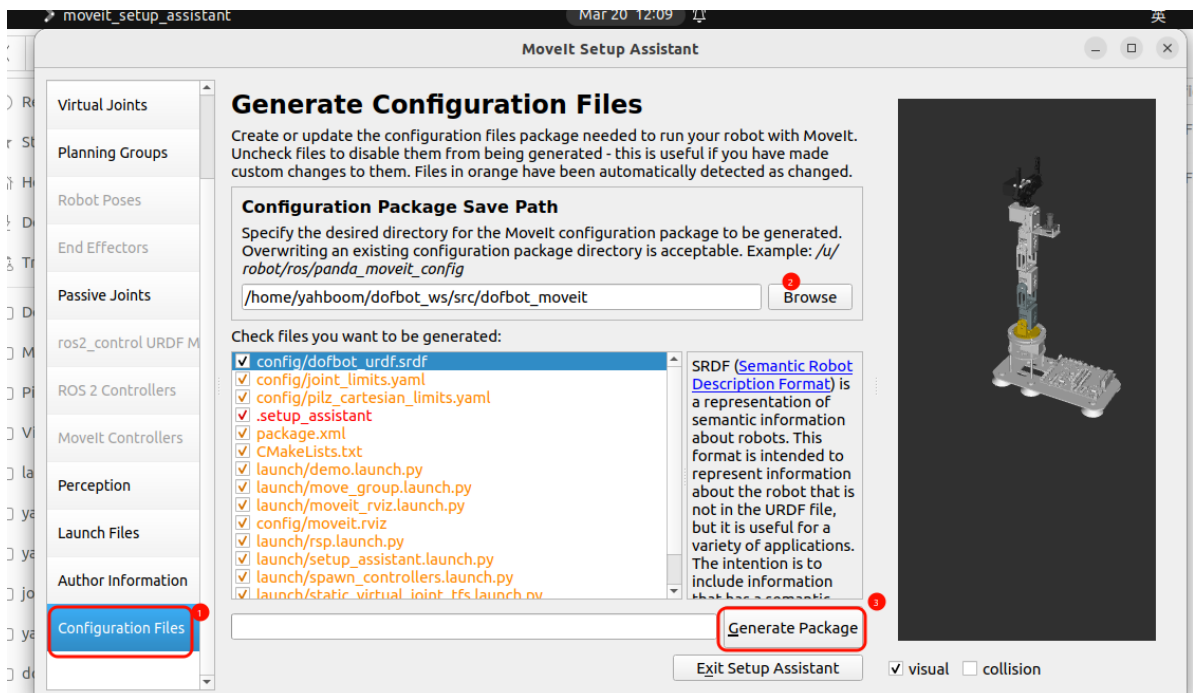
2.13. Author Information

Add author information for the purpose of generating the software package:

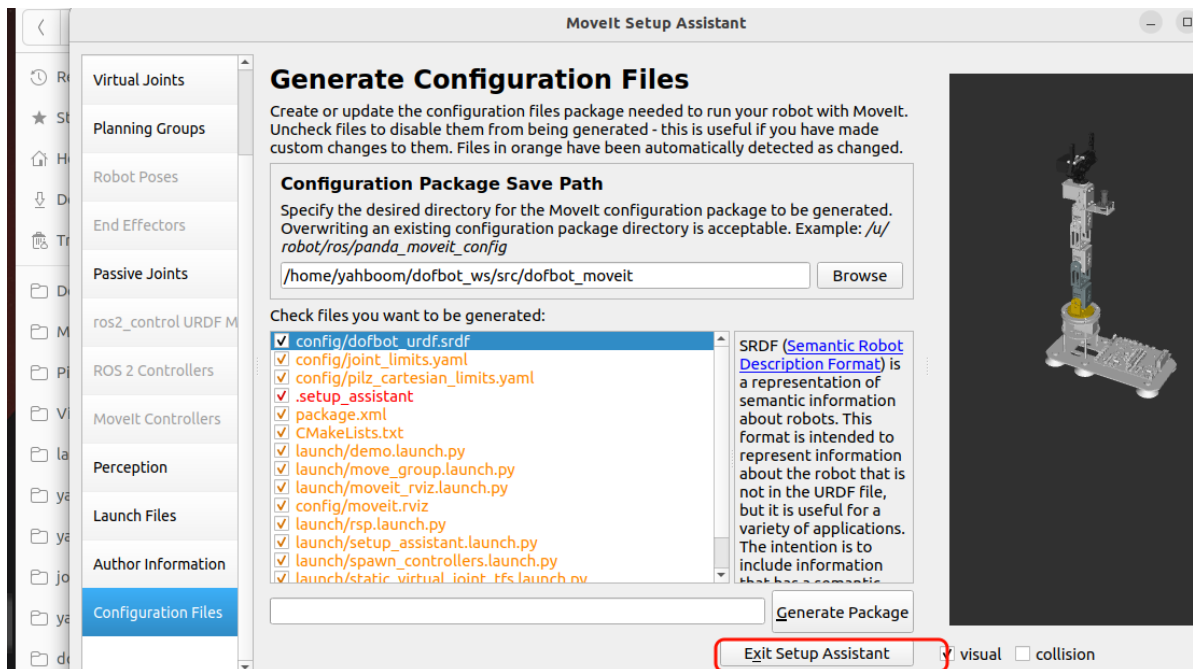


2.14. Generate Configuration

Generate the configuration files into the specified folder:



A "Confirm Package Update" dialog box will pop up; click **OK**, then click **Exit Setup Assistant** at the bottom.



3. Configuration Verification

The configuration files generated by the MoveIt Setup Assistant require minor modifications to resolve startup warnings and issues regarding the model failing to load.

4.1. Modifying Configuration Files

joint_limits.yaml

Modify the `joint_limits.yaml` file: change the maximum velocity and acceleration values for all joints to decimal numbers.



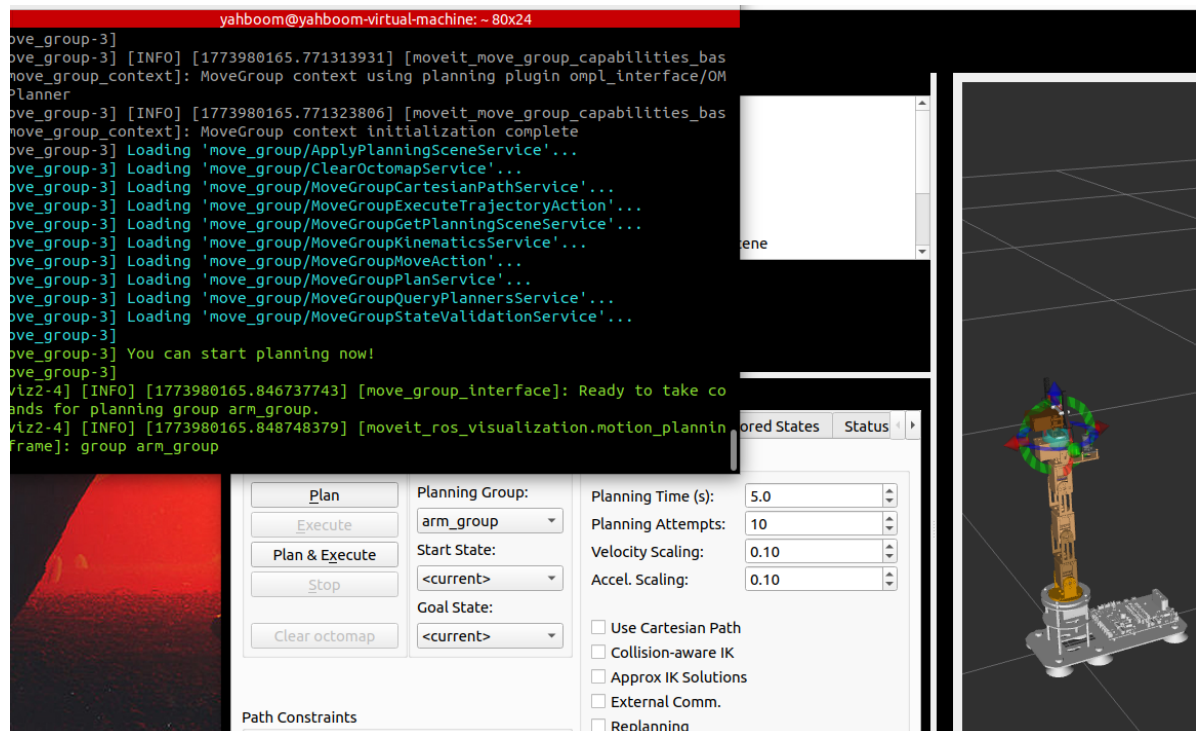
4.2. Building the Package

```
cd ~/dofbot_ws
colcon build --packages-select dofbot_moveit
source install/setup.bash
```

4.3. Launching MoveIt

```
ros2 launch dofbot_moveit demo.launch.py
```

Launching the simulation may take some time. Loading is complete once the message `You can start planning now!` appears in the terminal, or when a "trackball" (a newly added spherical marker) appears on the robotic arm.



References

MoveIt2 Humble: <https://moveit.picknik.ai/humble/index.html>

MoveIt2: <https://moveit.picknik.ai/main/index.html>